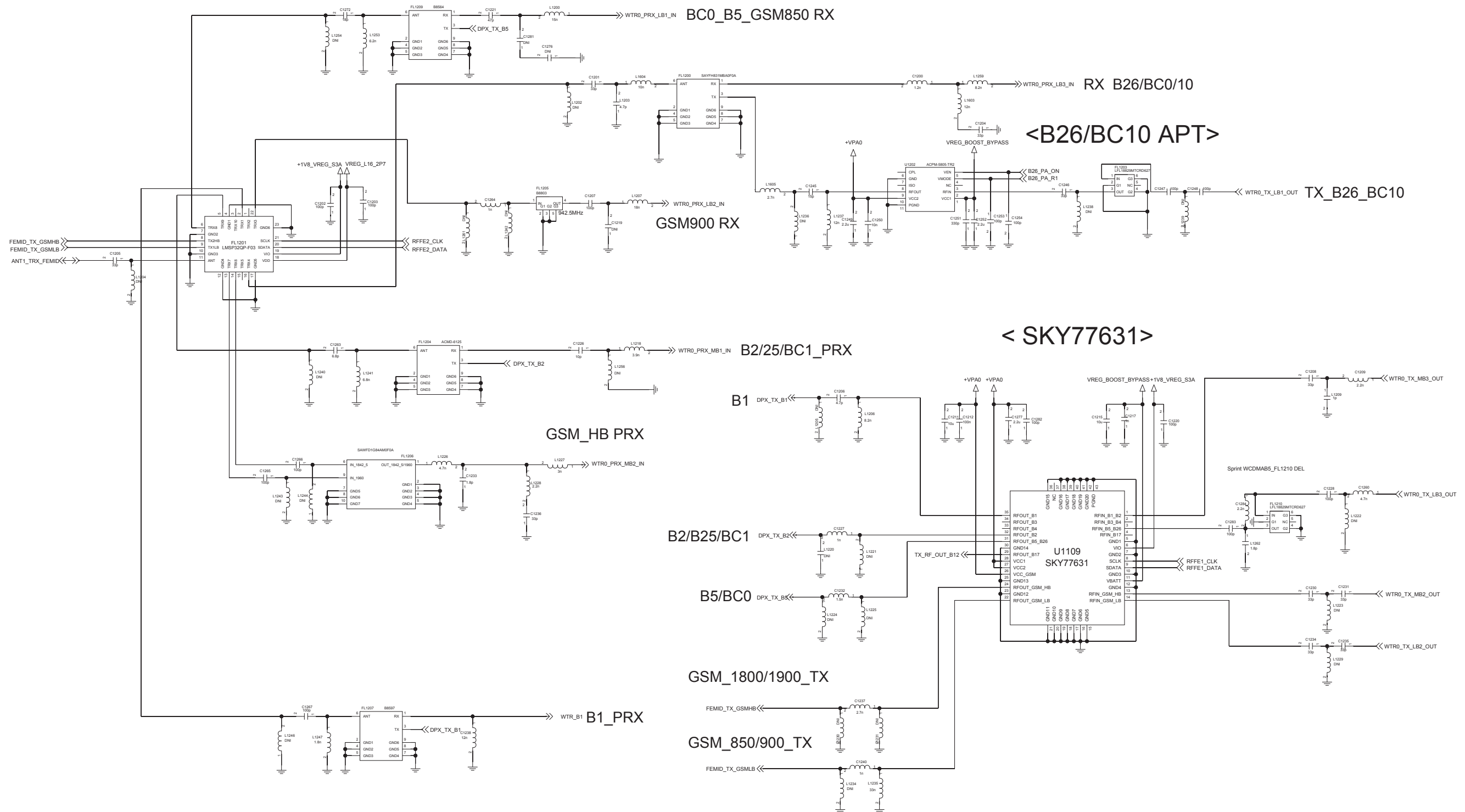
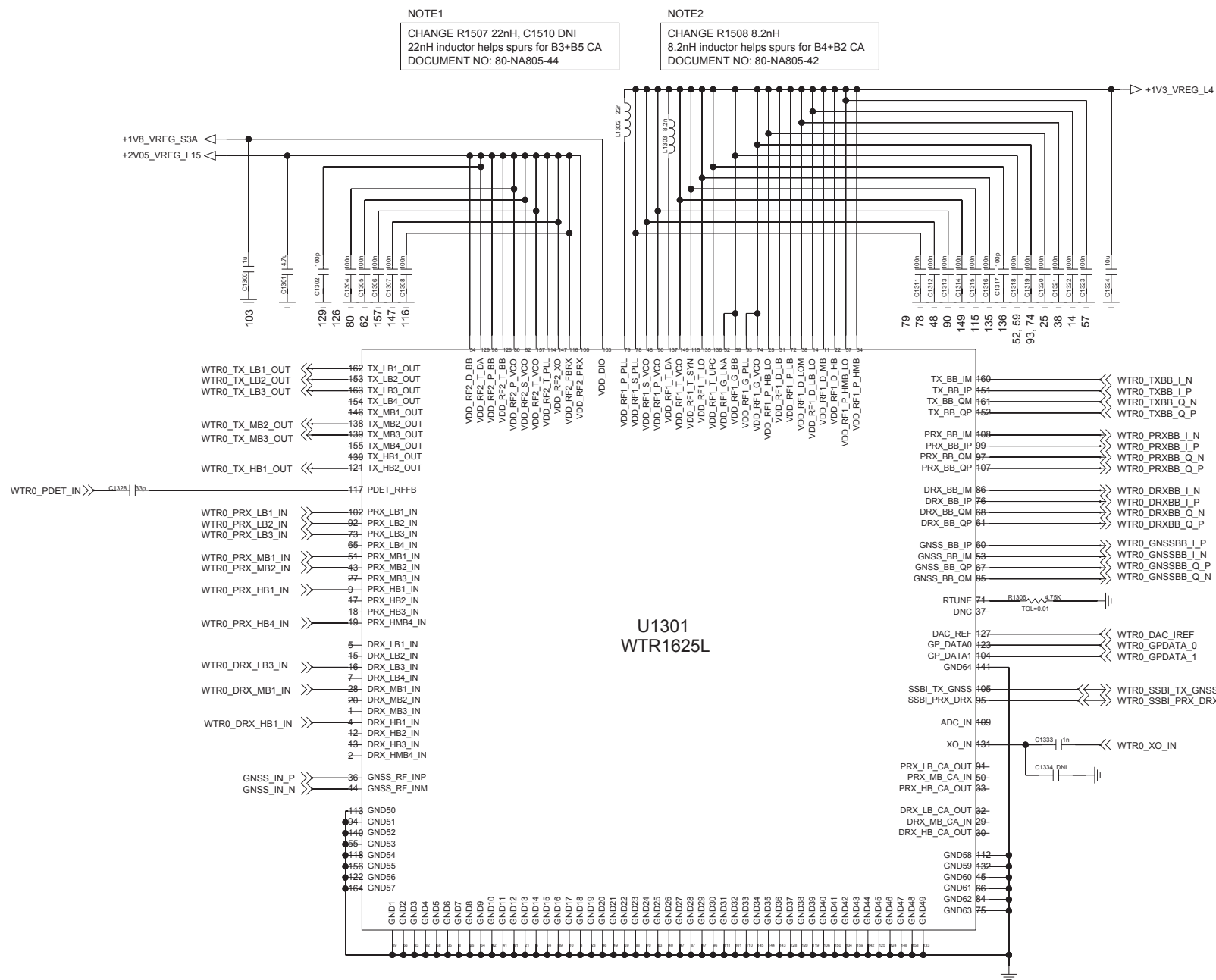


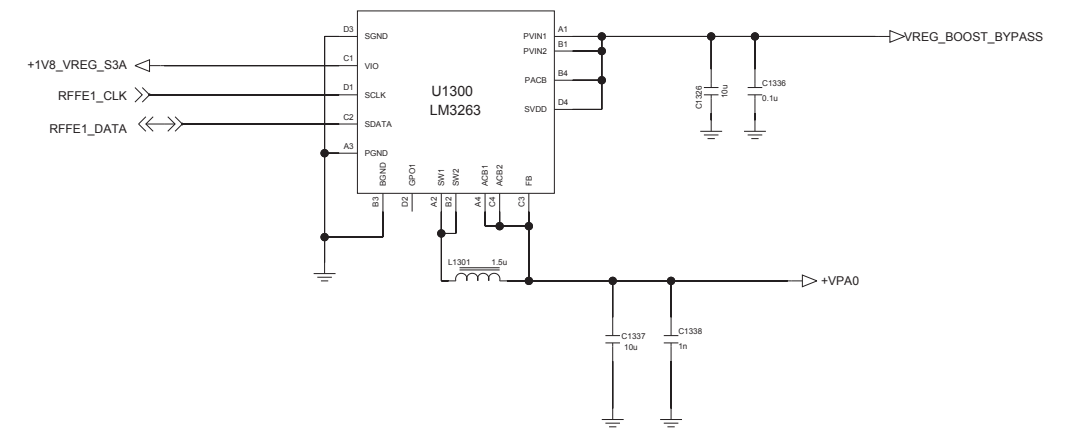


< 1-3-5-1_RFIC_CA_nCA_WTR1625 >

CAUTION: YOU MUST CHECK & MODIFY RF PORT AND GPIO ON YOUR SCHEMATICS.

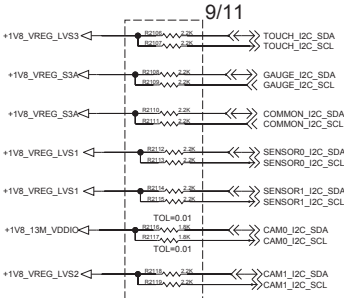


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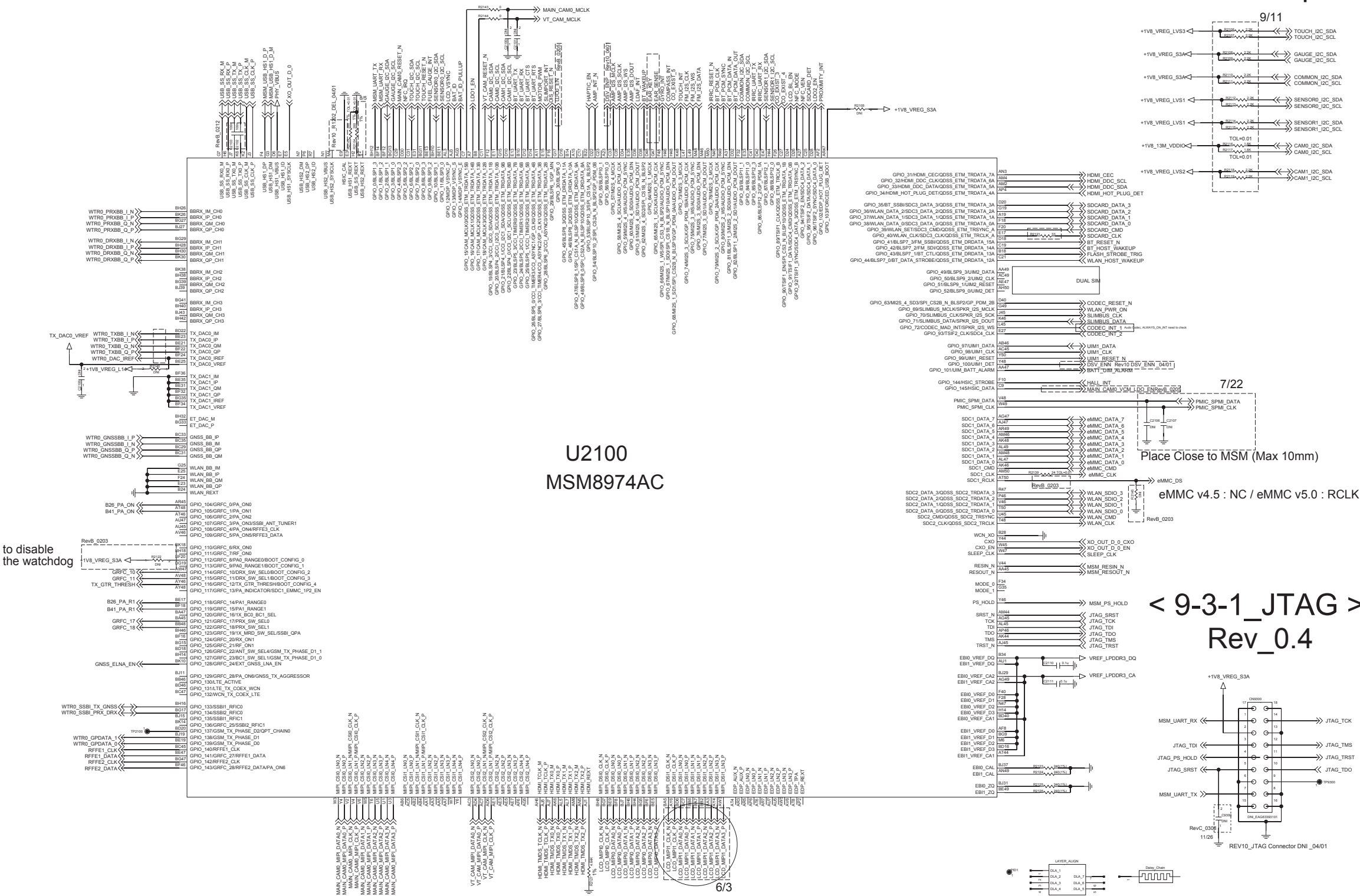


<2-1-2-2-1_MSM8974AB_Control_GPIO> Rev_0.1

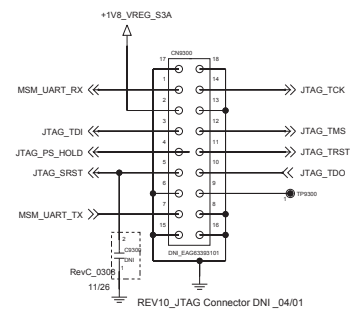
I2C Pull-Up



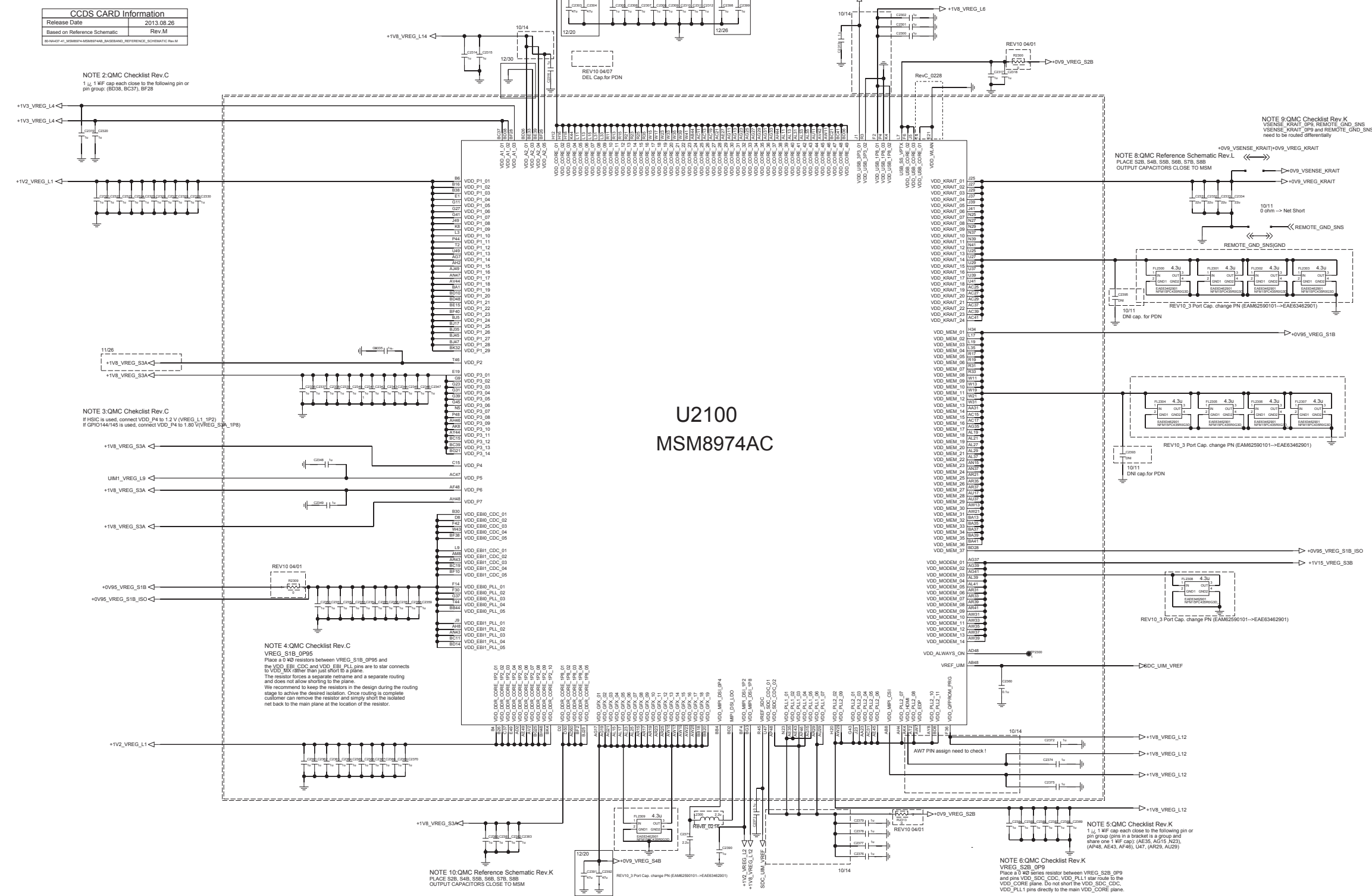
U2100
MSM8974AC



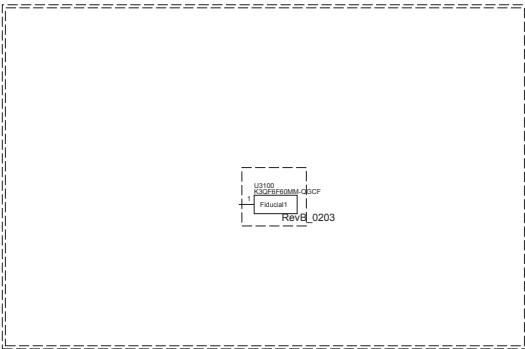
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Rev_0.4



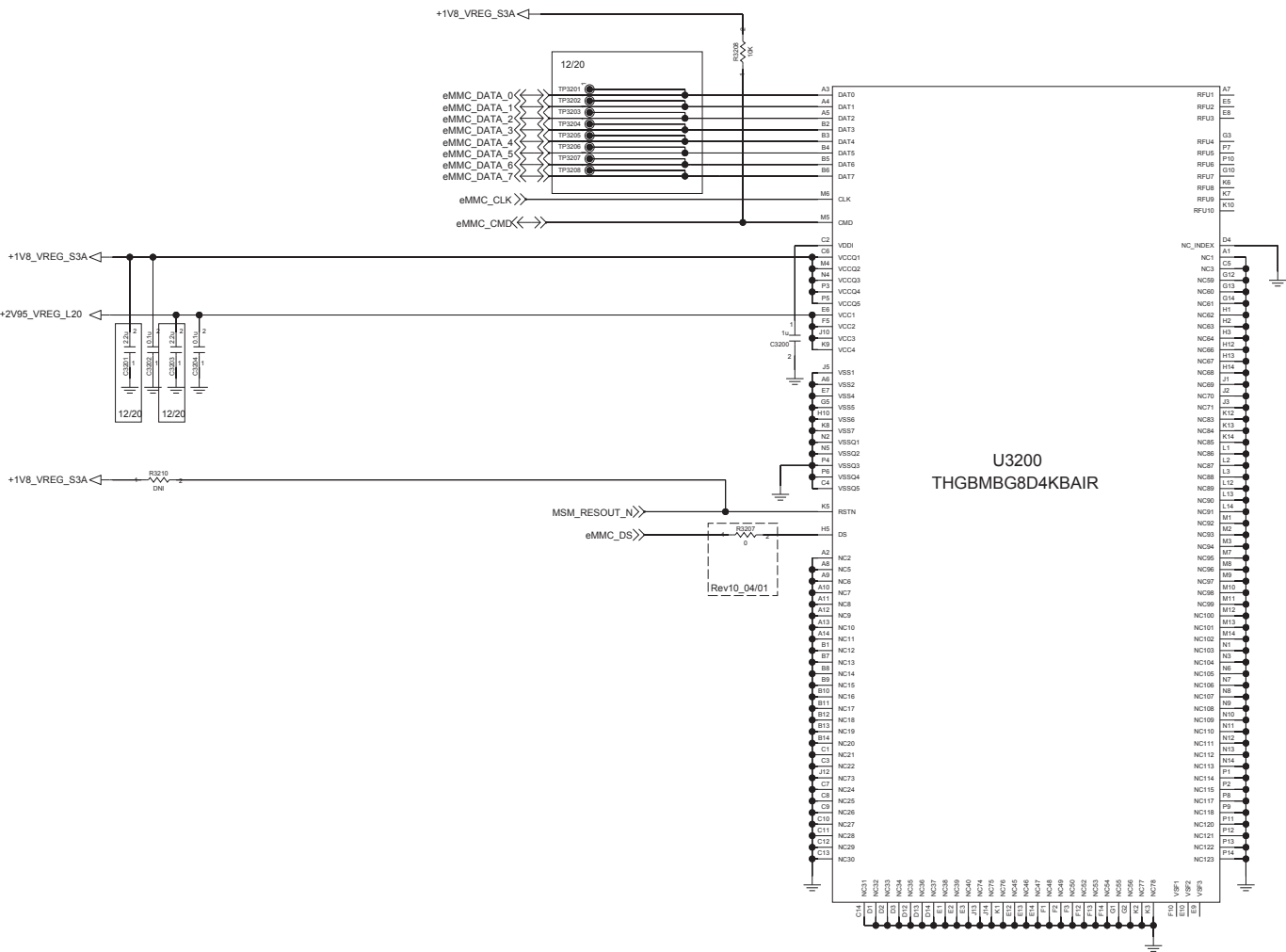
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Rev_0.1



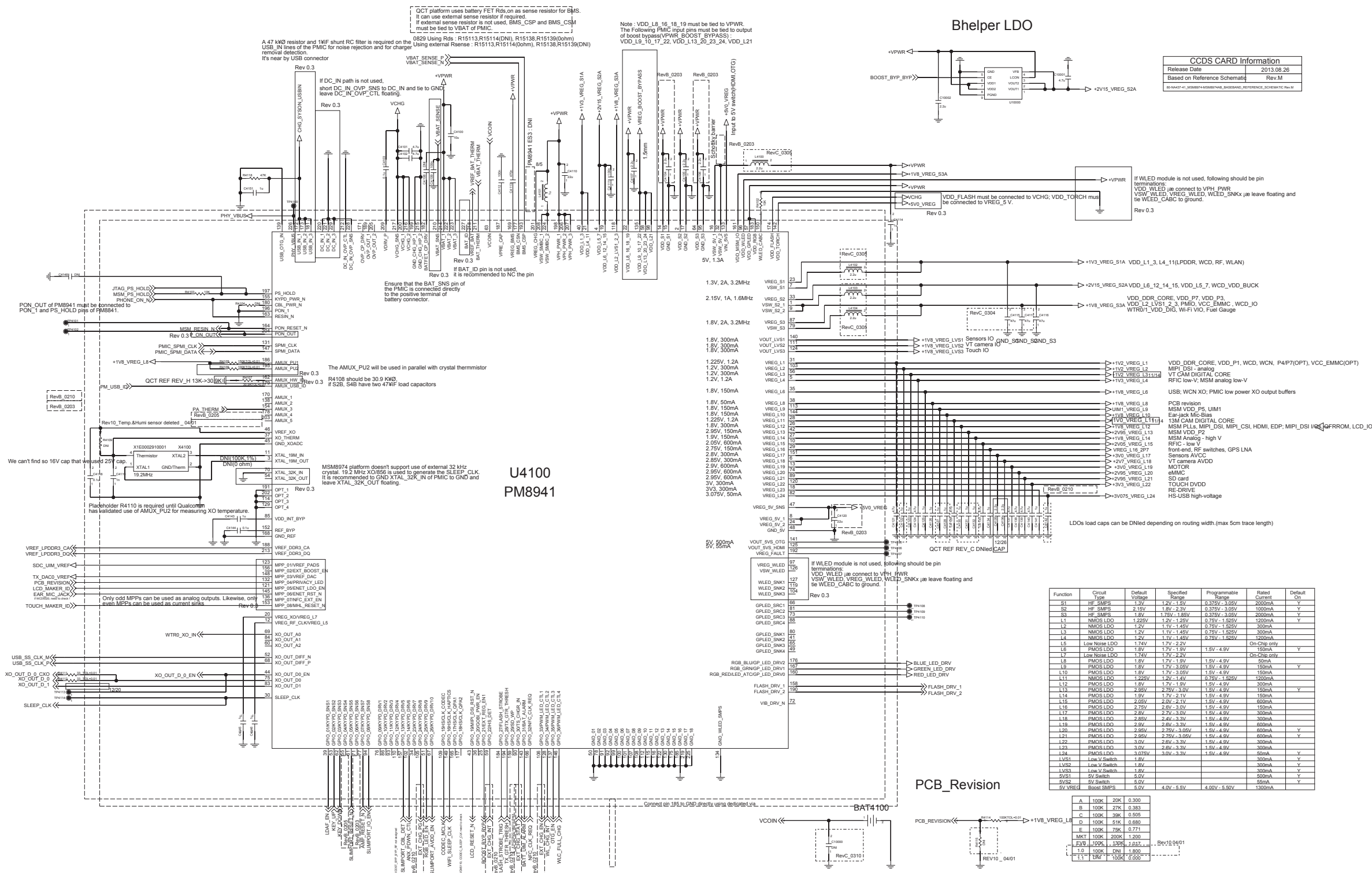
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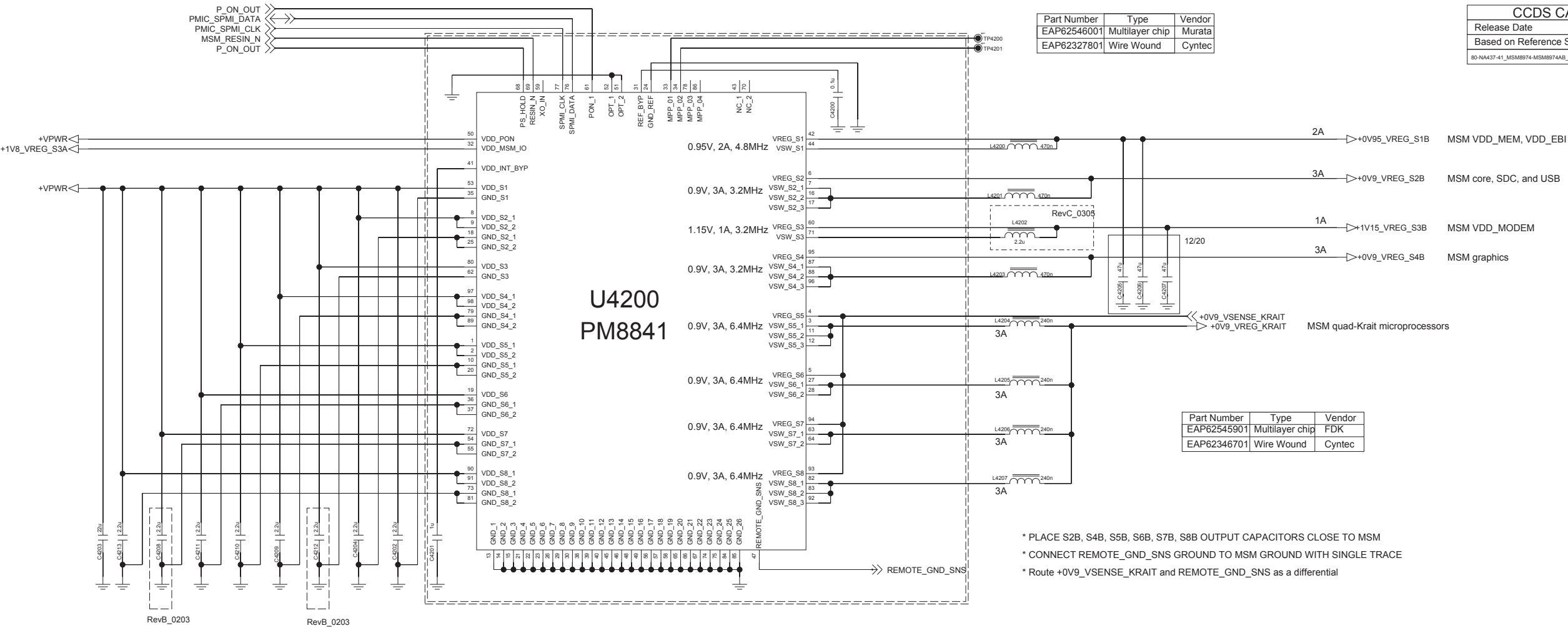
<3-2-2-3-1_eMMC_4_5_Toshiba_32G-Byte> Rev_0.4

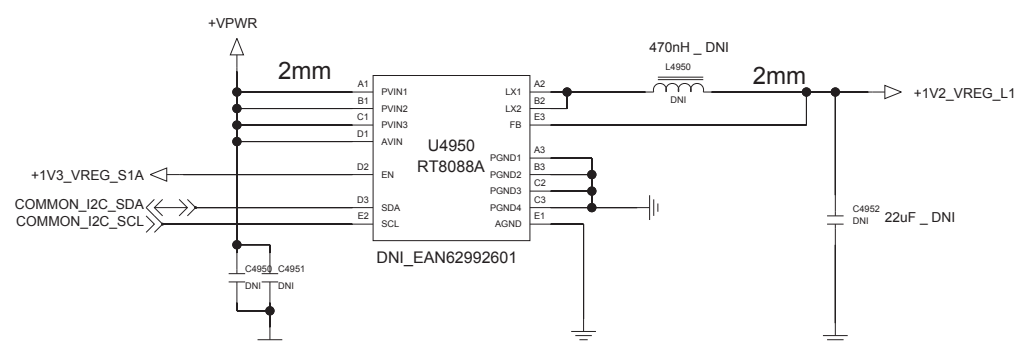


<4-1-2_PMIC_PM8941> Rev_0.3



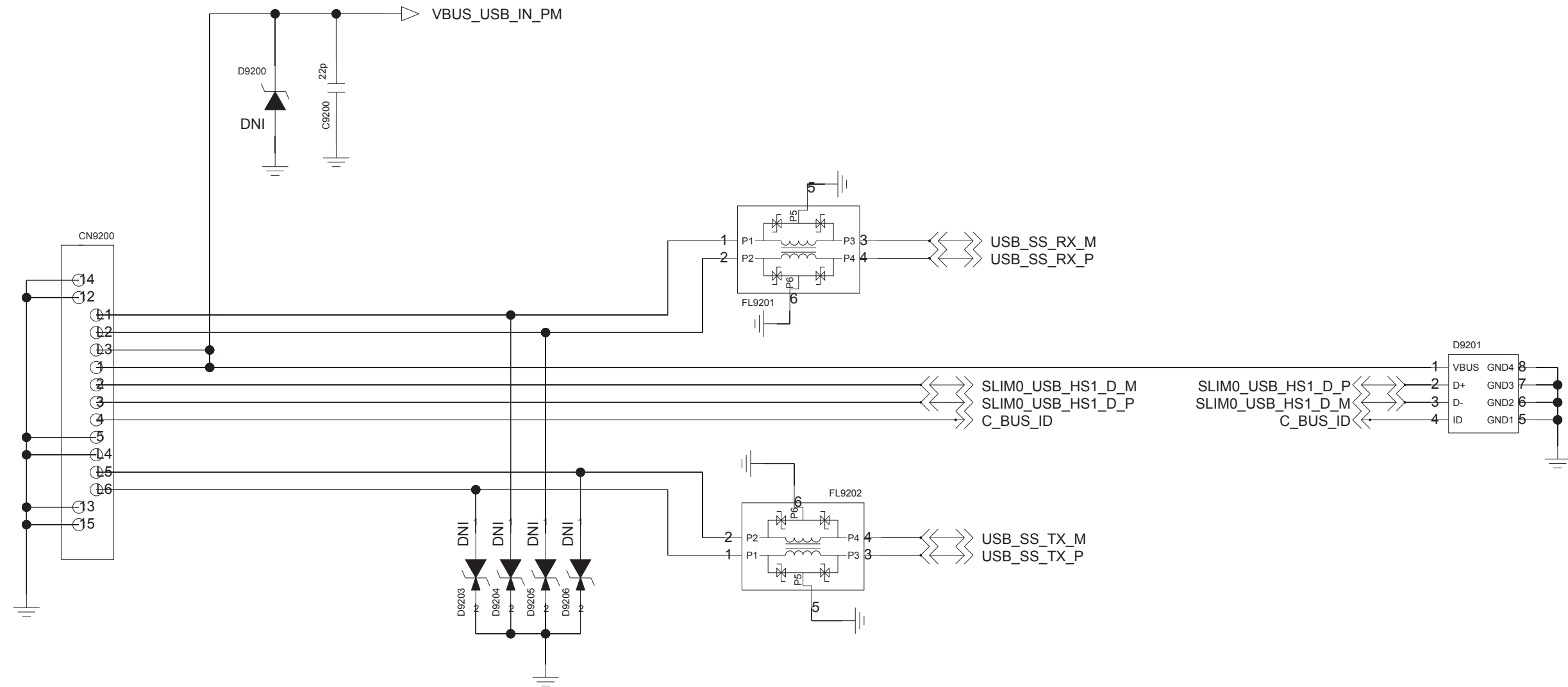
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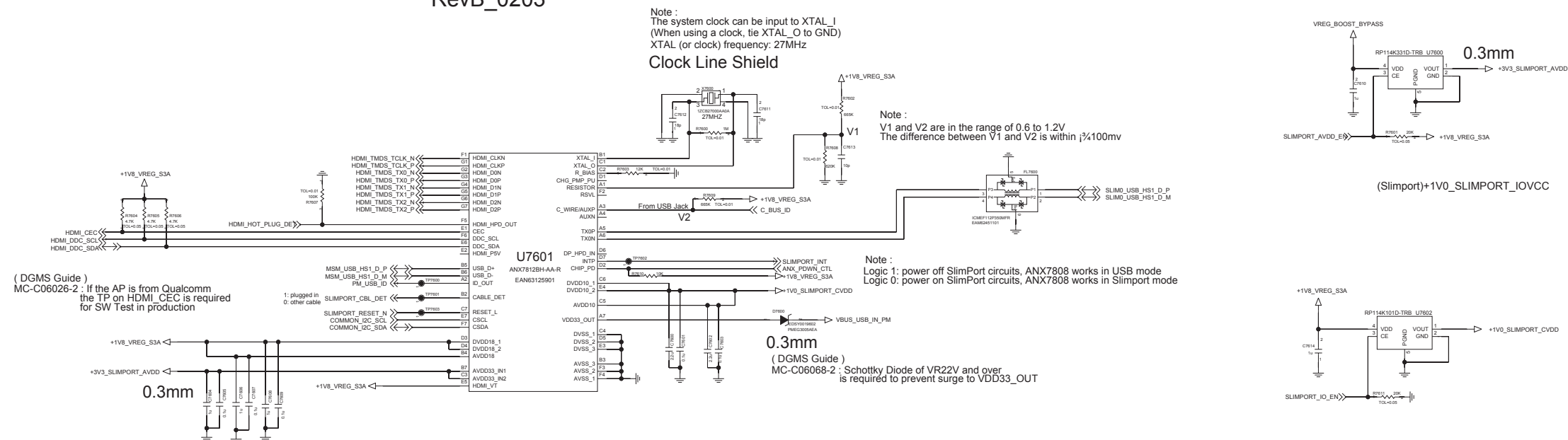
<9-2-2_Multi_IO_USB3_0> Rev_0.3



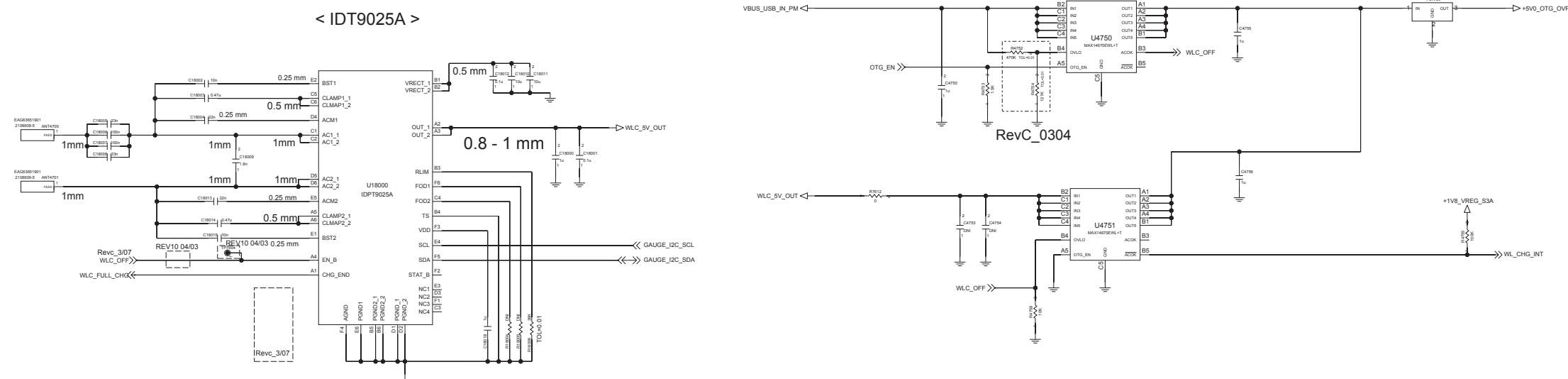
SLIMPORT/USB3.0 Rev_0.3

<7-6-2-1_SlimPort_ANX7812BH-AA-R>

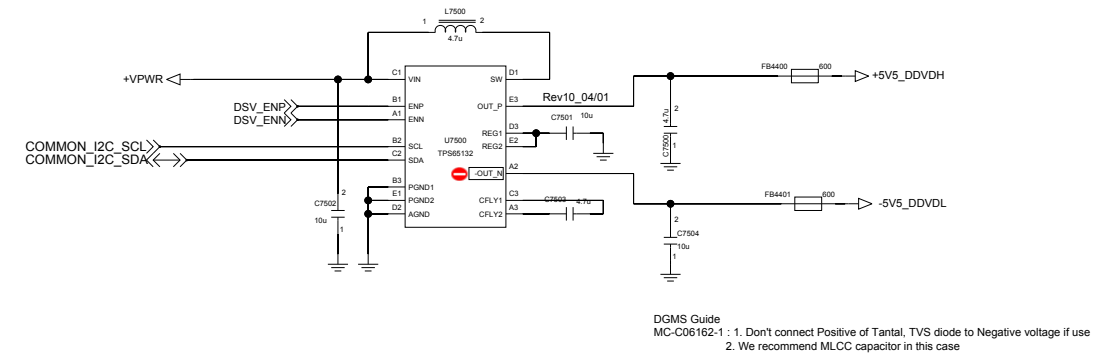
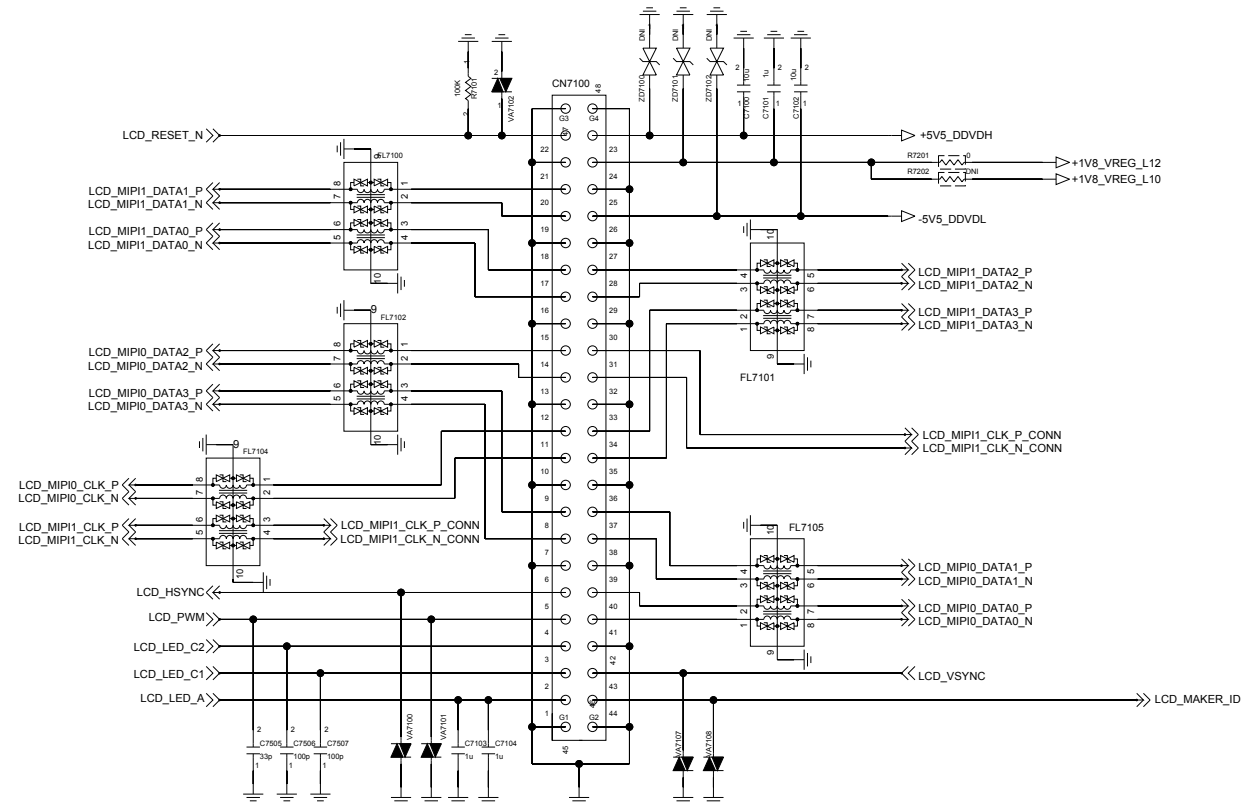
RevB_0203



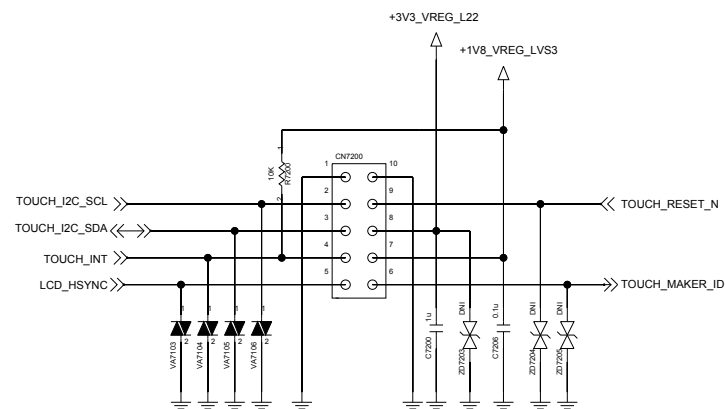
Wireless Charger <4-7-3-1_Wireless_Charger_MAX14670EWL_T>
Rev_0.3



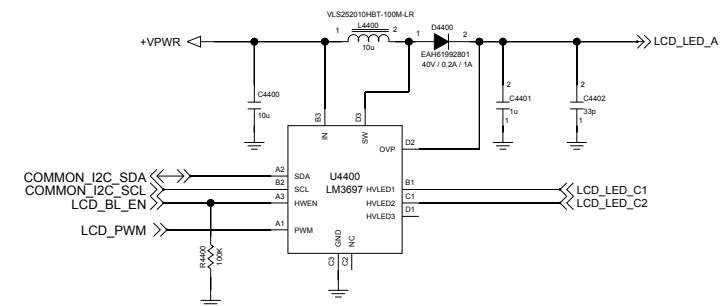
<7-1-8-4-2_Quad_LCD_5.5"> Rev_0.1

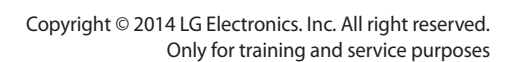


5.5" Touch

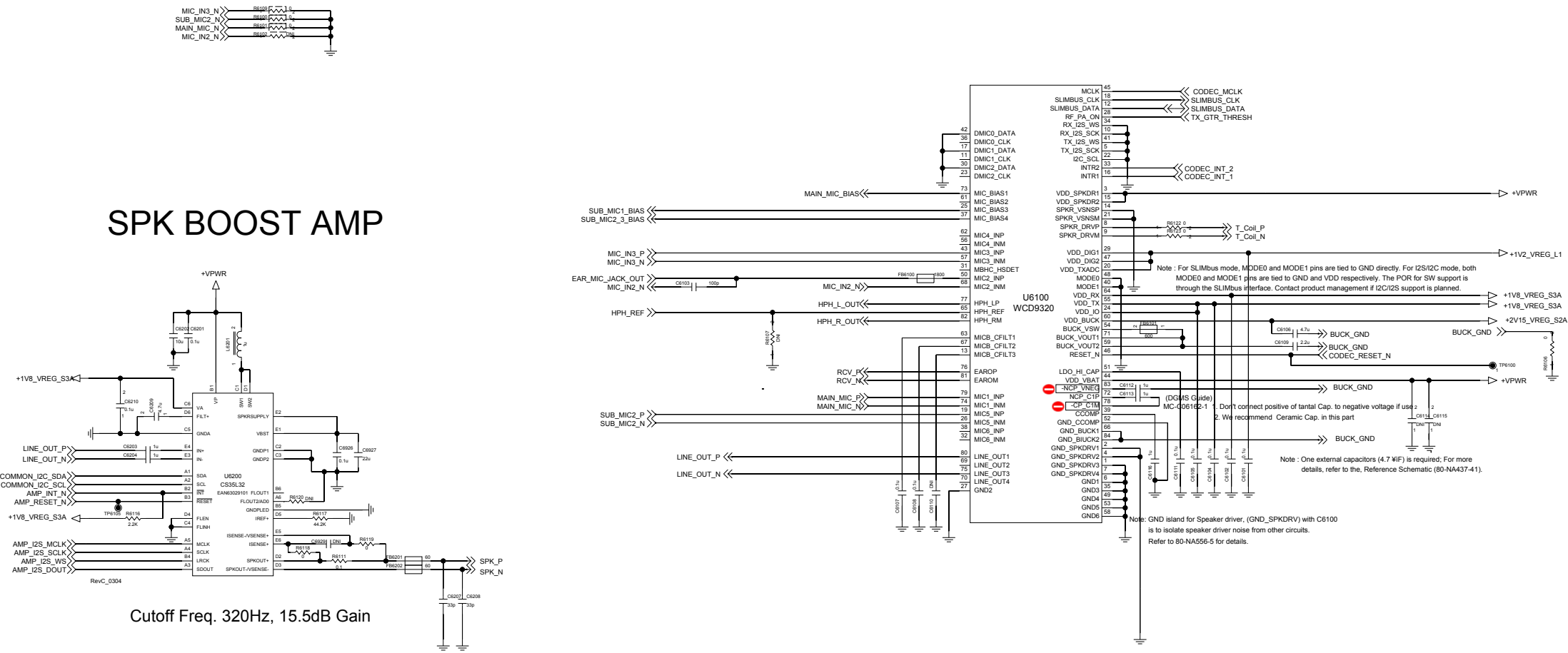


<4-4-4_LCD_BL_Booster_LM3697> Rev_0.1

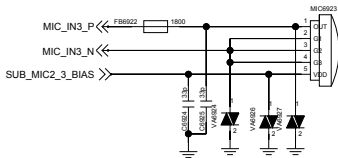




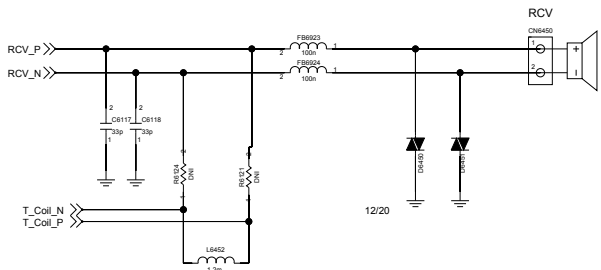
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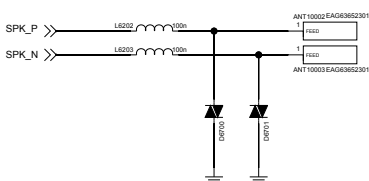
Sub_MIC2



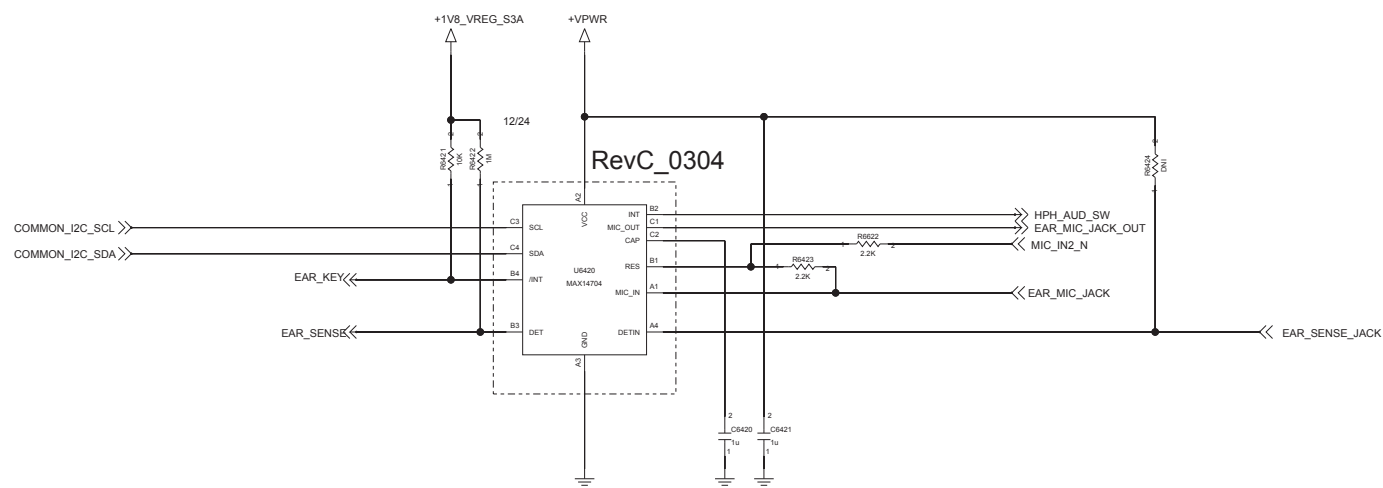
< 6-8-2_Receiver > Rev_0.4



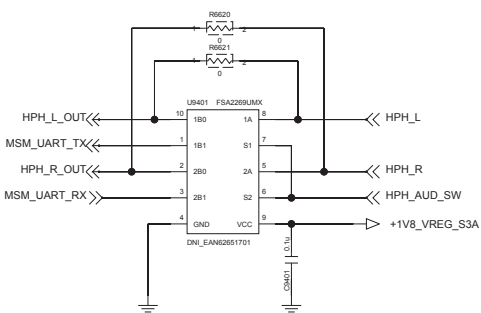
< 6-7-1_Speaker > Rev_0.4



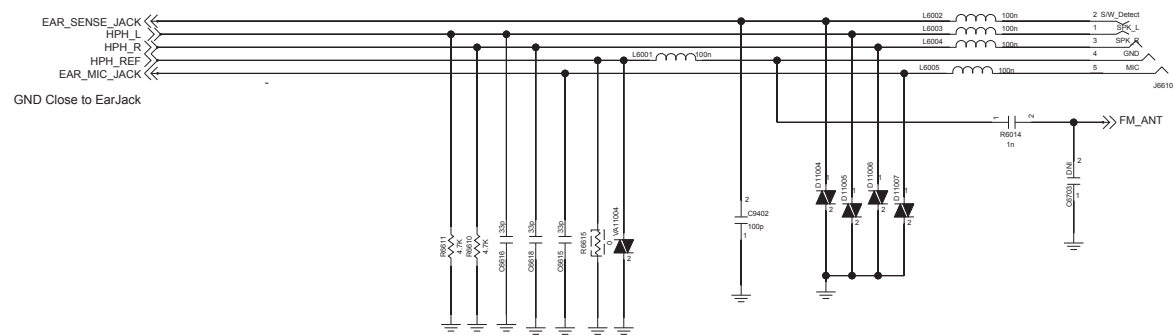
< Earjack detect IC >



<Earjack_Debug_Port >



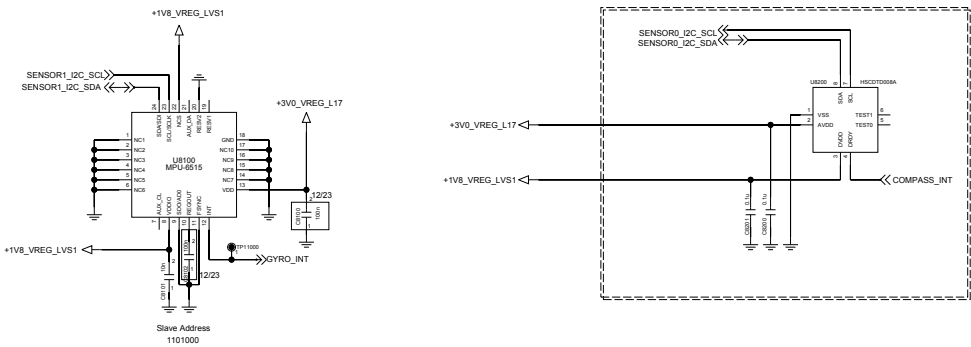
< 6-6-1_Earjack > Rev_0.5



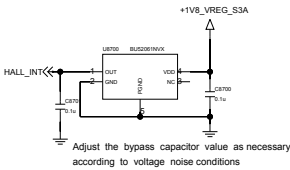
SENSOR/MOTOR/IRRRC/SIM/SD/LED

I2C	Devices
Sensor0	Compass, Temp+Humid+Pressure
Sensor1	Accel+Gyro, Pressure, UV, Prox+RGB+IR Gesture

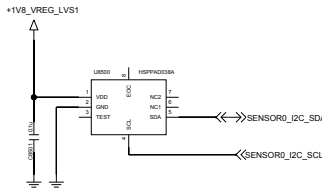
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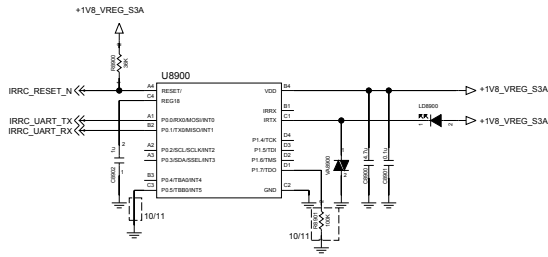
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<Pressure_HSPPAD038A >

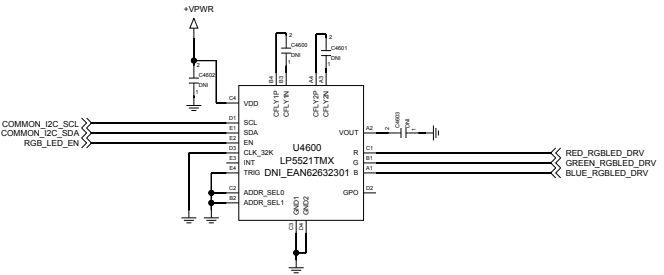


< 8-9-1-1_IRRC_MAXQ616V > Rev_0.4

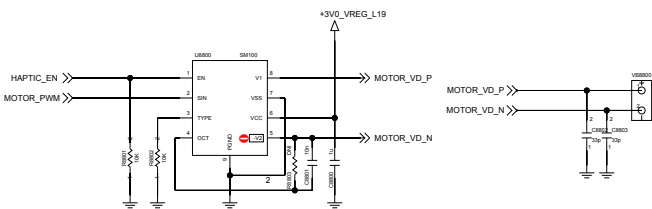


REV.1.0

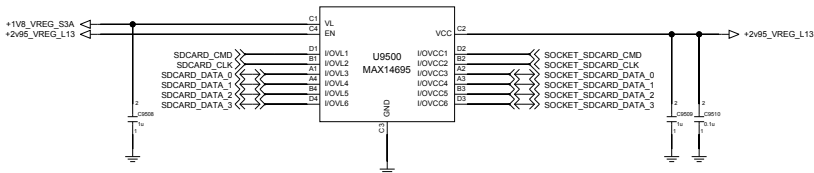
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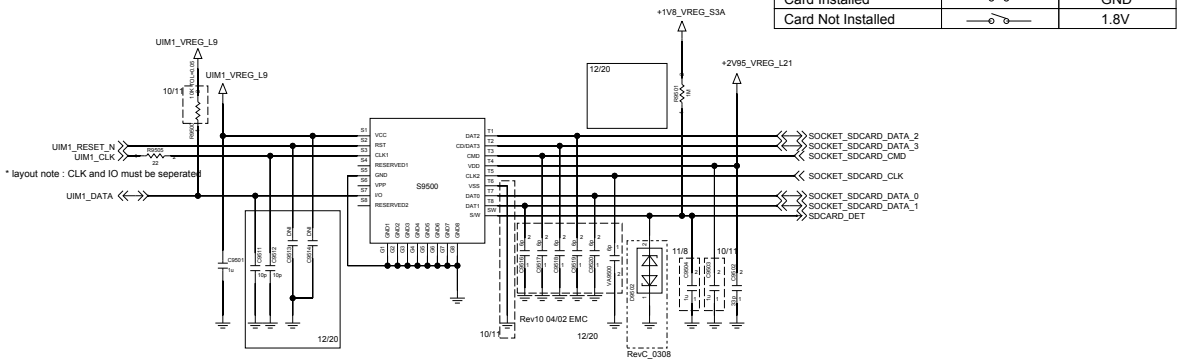
< 8-8-1-1_Haptic_SM100 >



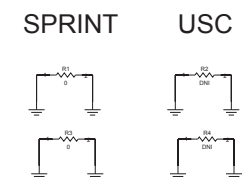
<LEVEL TRANSLATOR>



< 9-5-3_Micro_SD_Combo > Rev_0.3



Card Detect	SW	VSS(T6)	SD_CARD_DET
Card Installed	—○—	—○—	GND
Card Not Installed	—○—	—○—	1.8V

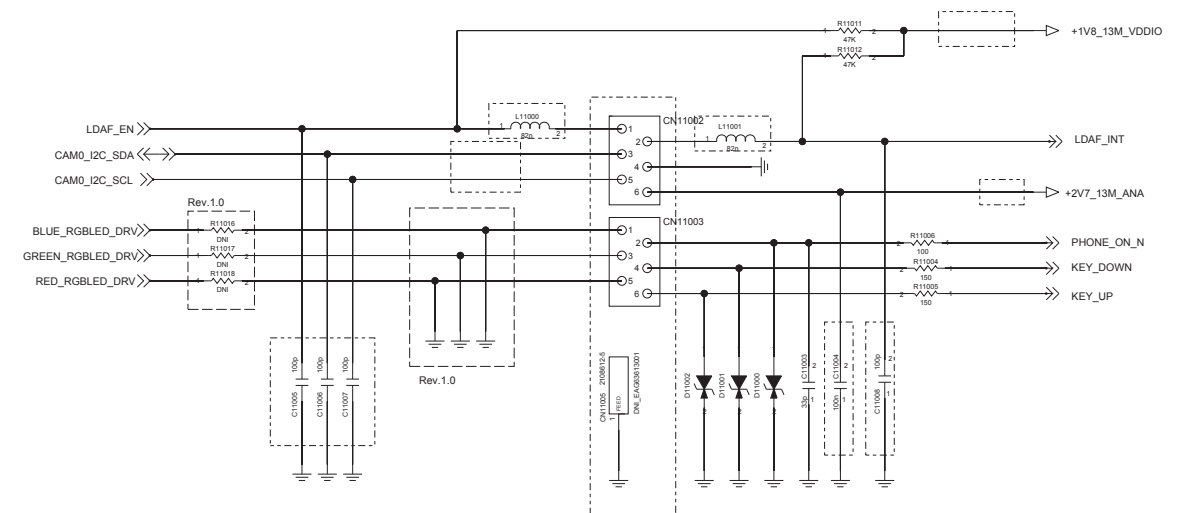
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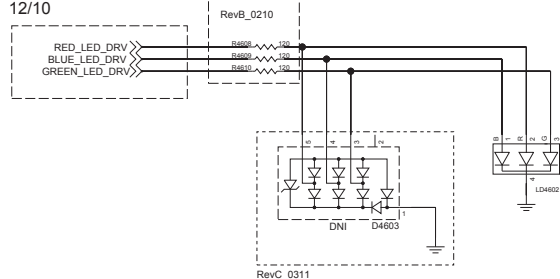
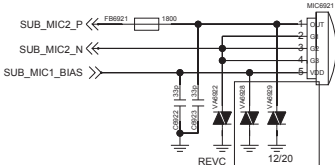
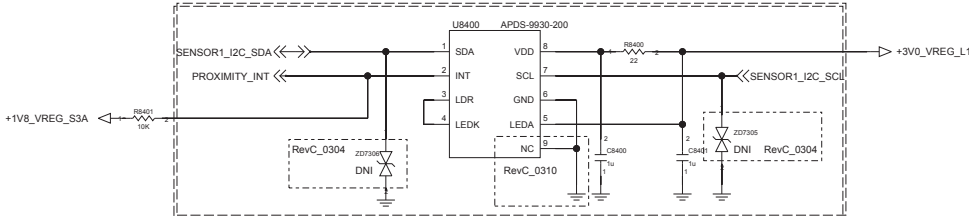
FPCB_CNT

I2C	Devices
Sensor0	Compass, Temp+Humid+Pressure
Sensor1	Accel+Gyro, Pressure, UV, Prox+RGB+IR Gesture

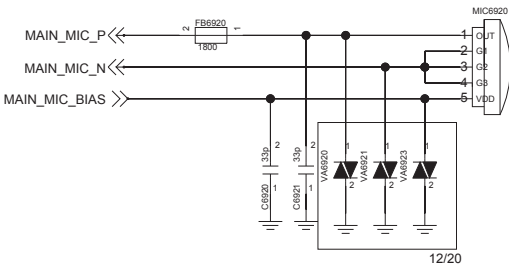
RevC_0302

Back_Key_LDFAF_FPCB



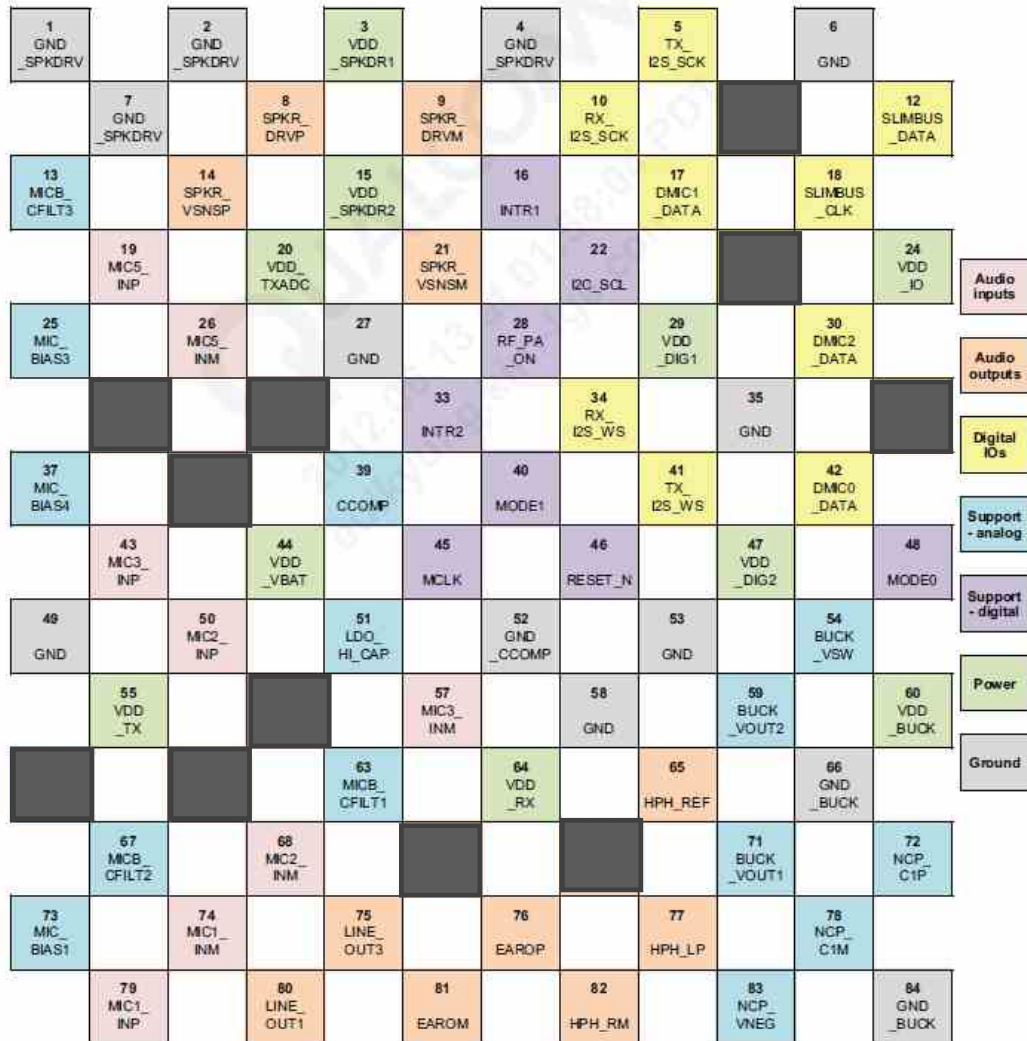


From previously Lower FPCB to Main PCB (11/13 now)



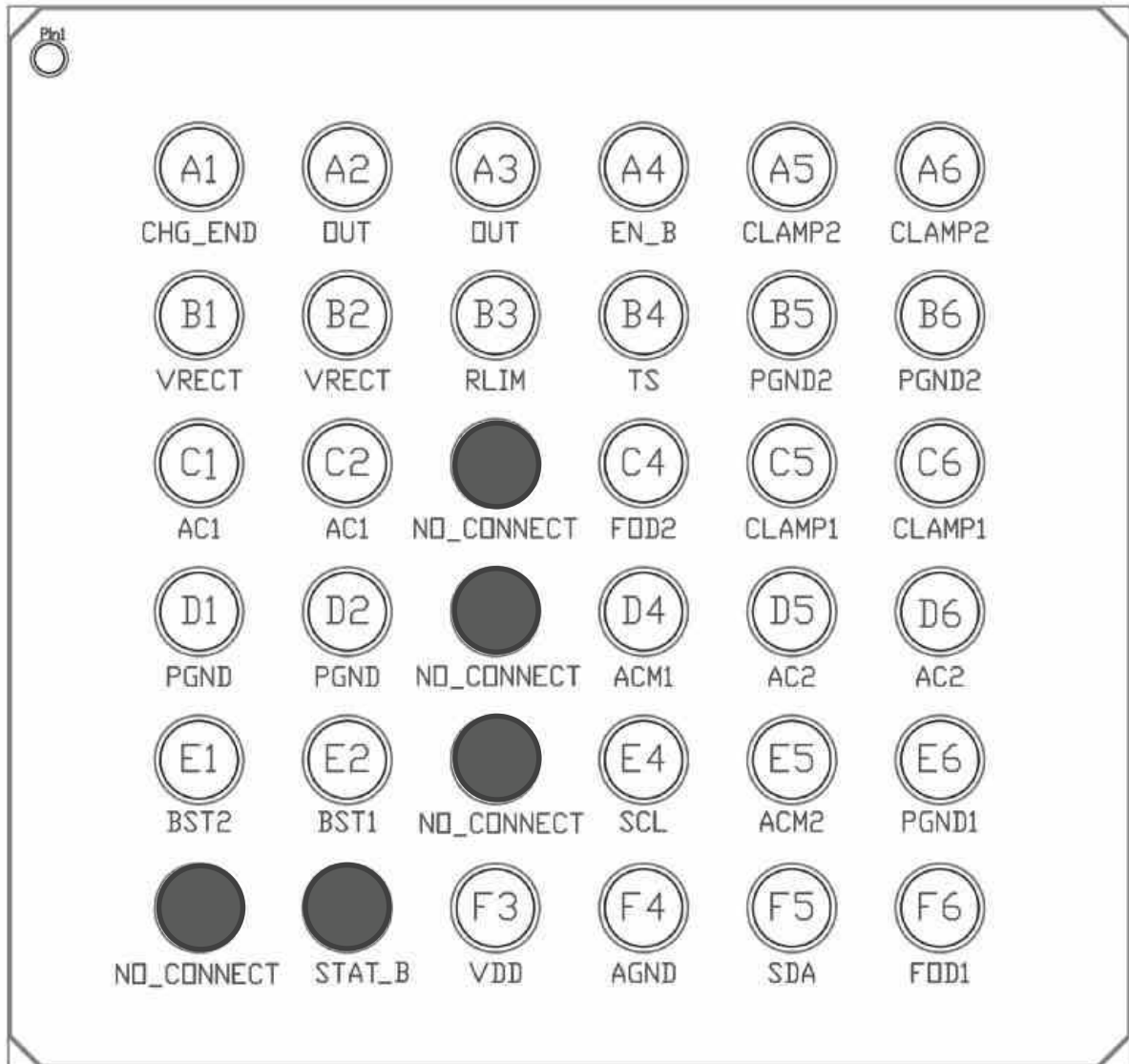
7. BGA PIN MAP

U6100_WCD9320_IC,Audio Codec(Top view)



Not Used

U18000_IC,Charger_IDTP9025A (Top View)



● Not Used

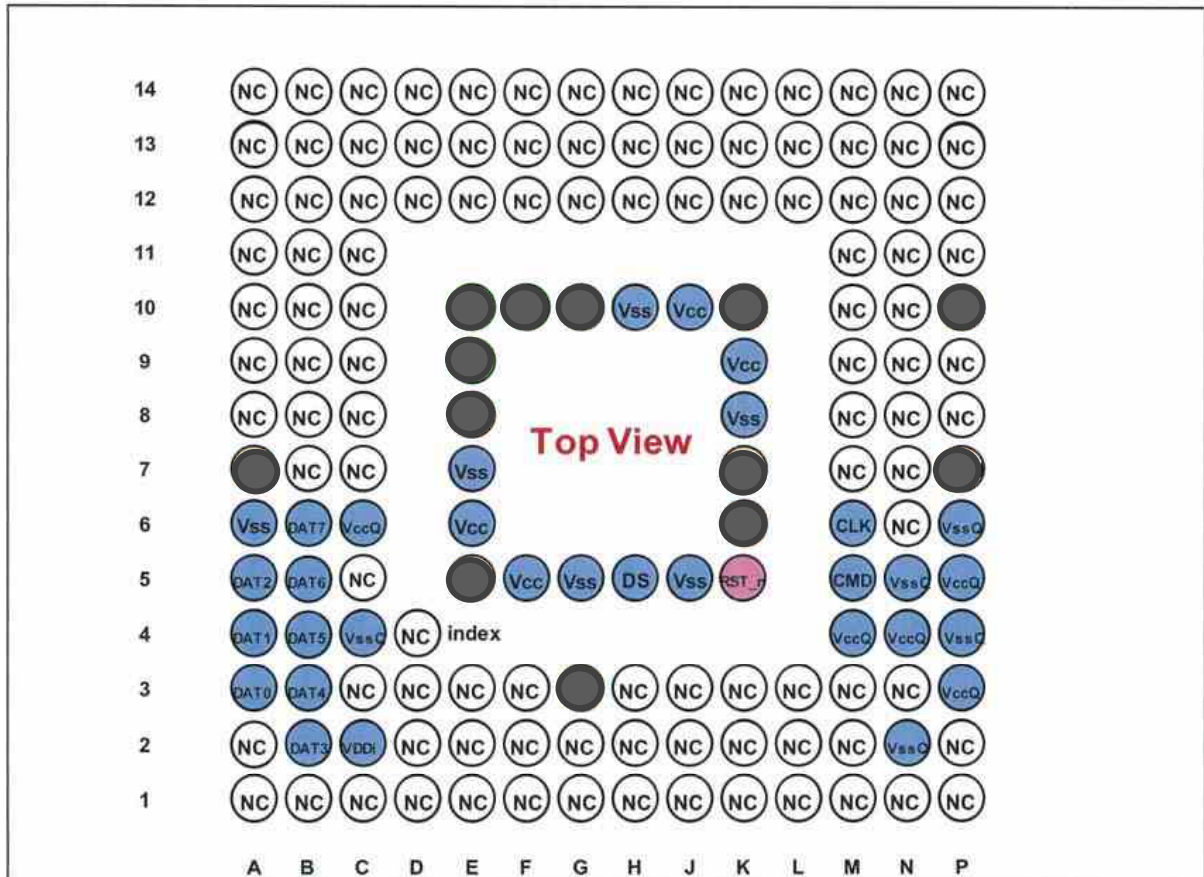
7. BGA PIN MAP

U4200_PM8841_PMIC(TOP view)

1 VDD _S5	2 VDD _S5	3 VSW _S5	4 VREG _S5	5 VREG _S6	6 VREG _S2	7 VSW _S2	8 VDD _S2	9 VDD _S2
10 GND _S5	11 VSW _S5	12 VSW _S5	13 GND	14 GND	15 GND	16 VSW _S2	17 VSW _S2	18 GND _S2
19 VDD _S6	20 GND _S5	21 GND	22 GND	23 GND	24 GND _REF		25 GND _S2	26 GND
27 VSW _S6	28 VSW _S6	29 GND	30 GND	31 REF _BYP	32 VDD _MSM_IO	33 MPP _01	34 MPP _02	35 GND _S1
36 GND _S6	37 GND _S6	38 GND	39 GND	40 GND	41 VDD _INT_BYP	42 VREG _S1		44 VSW _S1
45 GND	46 GND	47 REMOTE _GND_SNS	48 GND	49 DNC	50 VDD _PON	51 OPT_2	52 OPT_1	53 VDD _S1
54 GND _S7	55 GND _S7	56 GND	57 GND	58 DNC		60 VREG _S3	61 PON_1	62 GND _S3
63 VSW _S7	64 VSW _S7	65 GND	66 GND	67 GND	68 PS _HOLD	69 RESN _N		71 VSW _S3
72 VDD _S7	73 GND _S8	74 GND	75 GND	76 SPMI _DATA	77 SPMI _CLK		79 GND _S4	80 VDD _S3
81 GND _S8	82 VSW _S8	83 VSW _S8	84 GND	85 GND		87 VSW _S4	88 VSW _S4	89 GND _S4
90 VDD _S8	91 VDD _S8	92 VSW _S8	93 VREG _S8	94 VREG _S7	95 VREG _S4	96 VSW _S4	97 VDD _S4	98 VDD _S4
OUTPUT PWR MGT	IC1/F	MPPs and GPIOs	No Connection	Power	Ground			

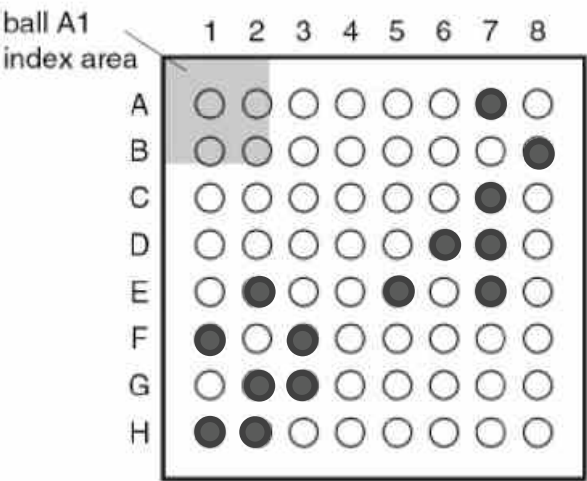
 Not Used

U3200_eMMC(TOP_view)



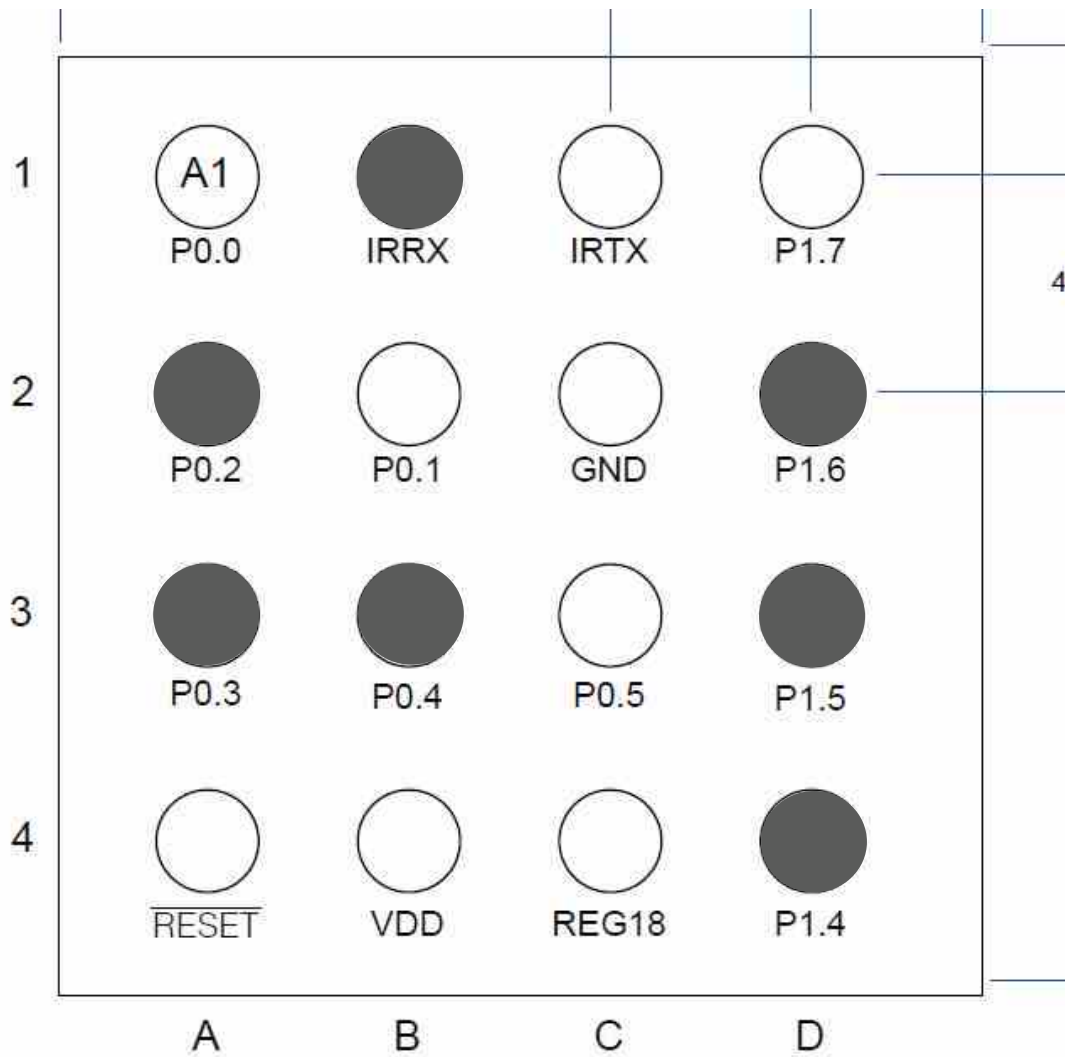
● Not Used

U5200_PN650/NFC Controller



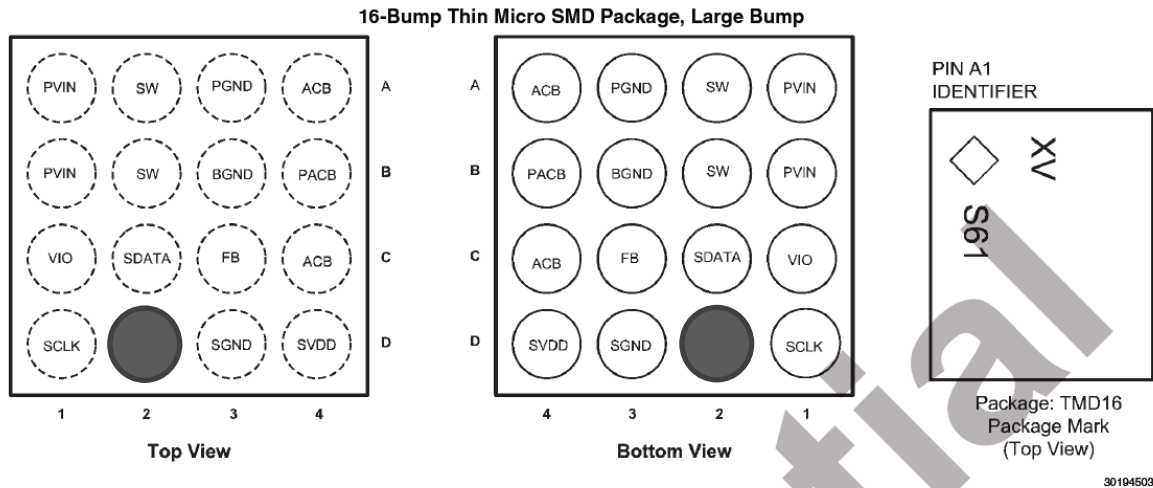
● Not Used

U8900_IC, Microprocessors (Bottom view)



● Not Used

U1300_IC,DC-DC Converter



Pin Descriptions

Pin #	Name	Description
A1	PVIN	Power Supply Voltage Input to the internal PFET switch and ACB.
B1		
C1	VIO	VIO functions as the RFFE interface reference voltage. VIO also functions as reset and enable input to the LM3263. Typically connected to voltage regulator controlled by RF or Baseband IC.
D1	SCLK	Digital control interface RFFE Bus clock input. Typically connected to RFFE master on RF or Baseband IC. SCLK must be held low when VIO is not applied.
A2	SW	Switching Node connection to the internal PFET switch and NFET synchronous rectifier.
B2		
C2	SDATA	Digital control interface RFFE Bus data input/output. Typically connected to RFFE master on RF or Baseband IC. SDATA must be held low when VIO is not applied.
D2	GPO1	General Purpose Output. Also used to reconfigure USID.
A3	PGND	Power Ground to the internal NFET switch.
B3	BGND	Active Current assist and Analog Bypass Ground and Digital Ground.
C3	FB	Feedback Analog Input. Connect to the output at the output filter capacitor.
D3	SGND	Signal Analog Ground (Low Current).
A4	ACB	Active Current assist and analog Bypass output. Connect to the output at the output filter capacitor.
C4		
B4	PACB	ACB Power Supply Input.
D4	SVDD	Analog Power Supply Voltage.



U7601_IC,Signal Bridge

	1	2	3	4	5	6	7	
A	RESISTOR	ID_OUT	C-WIRE /AUXP (AUX_HPD)		TX0P	TX0N	VDD33_OUT	A
B	XTAL_I	CABLE_DET	AVSS	AVDD18	USB_D+	USB_D-	AVDD33_IN	B
C	XTAL_O	R_BIAS	AVDD33_IN	DVSS	AVDD10	DVDD10	RESET_L	C
D		CHIP_PD	DVDD18	DVDD18	DVSS		INTP	D
E	CEC		DVSS	DVDD10	HDMI_VT	DDC_SDA	CSCL	E
F	HDMI_CLKN		AVSS	AVSS	HDMI_HPD_OUT	DDC_SCL	CSDA	F
G	HDMI_CLKP	HDMI_D0N	HDMI_D0P	HDMI_D1N	HDMI_D1P	HDMI_D2N	HDMI_D2P	G
	1	2	3	4	5	6	7	

 Not Used

7. BGA PIN MAP

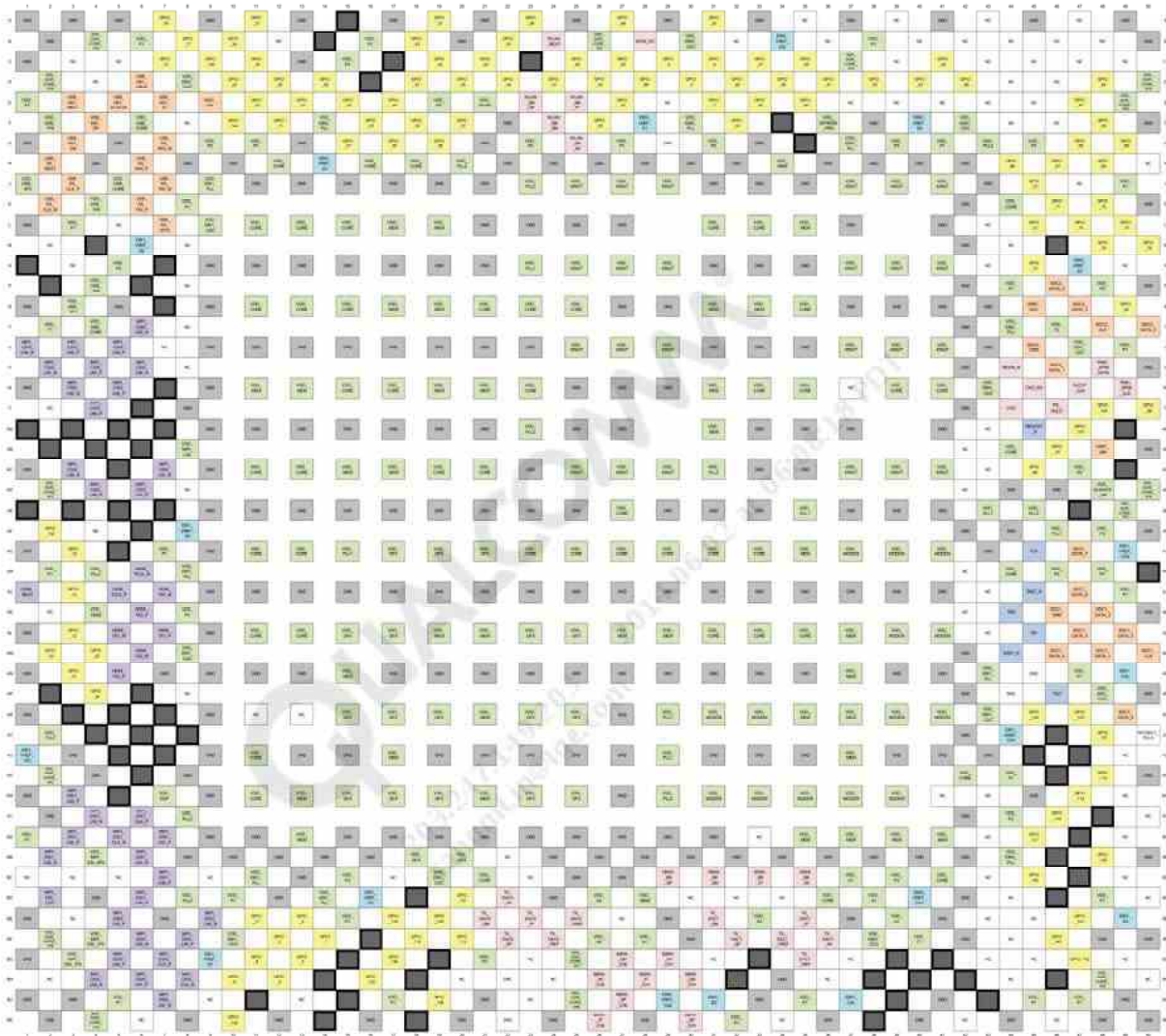
U21000_BCM4339,WiFi (Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	
A		NO CONNECT	NO CONNECT	NO CONNECT		NO CONNECT	NO CONNECT		NO CONNECT	NO CONNECT	NO CONNECT		A
B	SR_PVSS	SR_VLX	WL_REG_ON	LPO_IN	GPIO_3	GPIO_0	HSIC_DATA	HSIC_STROBE		SDIO_DATA_0	SDIO_CLK	SDIO_CMD	B
C	SR_VDDBATPSV	SR_VDDBATASV	PMU_AVSS		GPIO_4		WL_VDDC	HSIC_AVDD12PLL	HSIC_DVDD12	SDIO_DATA_1	SDIO_DATA_3	WL_VDDC	C
D	LDO_VDD1P5	VOUT_CLDO	BT_REG_ON		GPIO_5		VSSC	HSIC_AGN0PLL	VDDIO_5D	SDIO_DATA_2	VSSC		D
E	VOUT_3P3	VOUT_LN1D0	VSSC			BT_UART_CTS	VDDIO_RF	VSSC		RF_SW_CTRL_8	RF_SW_CTRL_3	RF_SW_CTRL_2	E
F	VOUT_BTLD02P5	LDO_VDDBATSV	VDDIO		RF_SW_CTRL_9	BT_UART_RTS	BT_UART_TXD				RF_SW_CTRL_1	RF_SW_CTRL_0	F
G	BT_PCM_IN	BT_PCM_CLK		WL_VDDC	WL_VDDC	BT_UART_RXD	RF_SW_CTRL_7	WL_VDDC		BBPLL_AV5	WRF_XTAL_GND1P2	BBPLL_AVDD1P2	G
H		BT_PCM_SYNC		BT_VDDIO	BT_VDDC	BT_I2S_WS	WRF_GPIO_OUT	WRF_WL_LNLD0M_VDD1P5		WRF_VCO_GND	WRF_XTAL_VDD1P5	WRF_XTAL_IN	H
J	FM_AUDIOVDD1P2	BT_HOST_WAKE	BT_PCM_OUT	BT_VDDC	VSSC	BT_I2S_CLK		WRF_BUCK_GND1P5	WRF_MMIO_GND1P2	WRF_PFD_GND1P2	WRF_CP_GND	WRF_XTAL_OUT	J
K		FM_AUDIOVSS	BT_DEV_WAKE	VSSC	BT_I2S_DI	BT_I2S_DO	WRF_AFE_GND1P2	WRF_IO_GND1P2_2	WRF_SYNTH_VBAT_VDD3P3	WRF_MMIO_VDD1P2	WRF_PFD_VDD1P2	WRF_XTAL_VDD1P2	K
L		FM_PLLVDD1P2	FM_PLLVSS	BT_PVDD1P2	BT_PLLVSS	BT_PVSS	WRF_RX2G_GND1P2	WRF_TX_GND1P2	WRF_PADREV_VBAT_VDD3P3	WRF_PADREV_VBAT_GND3P3	WRF_LO_GND1P2_2	WRF_RXSG_GND1P2	L
M	FM_VCOVSS	FM_LNAVSS	BT_VCOVSS	BT_PLLVDD1P2	BT_PAVSS		WRF_LNA_2G_GND1P2	WRF_PA_VBAT_GND3P3_4	WRF_PA_VBAT_GND3P3_3	WRF_PA_VBAT_GND3P3_2	WRF_PA_VBAT_GND3P3_1	WRF_LNA_5G_GND1P2	M
N	FM_LNAVCOVDD1P2	FM_RFIN	BT_VCOVDD1P2	BT_LNAVDD1P2	BT_RF	BT_PAVDD2P5	WRF_RFIN_2G	WRF_RFOUT_2G	WRF_PA2G_VBAT_VDD3P3	WRF_PA5G_VBAT_VDD3P3	WRF_RFOUT_5G	WRF_RFIN_5G	N
	1	2	3	4	5	6	7	8	9	10	11	12	



Not Used

U2100- MSM8974AC (TOP view)



 Not Used

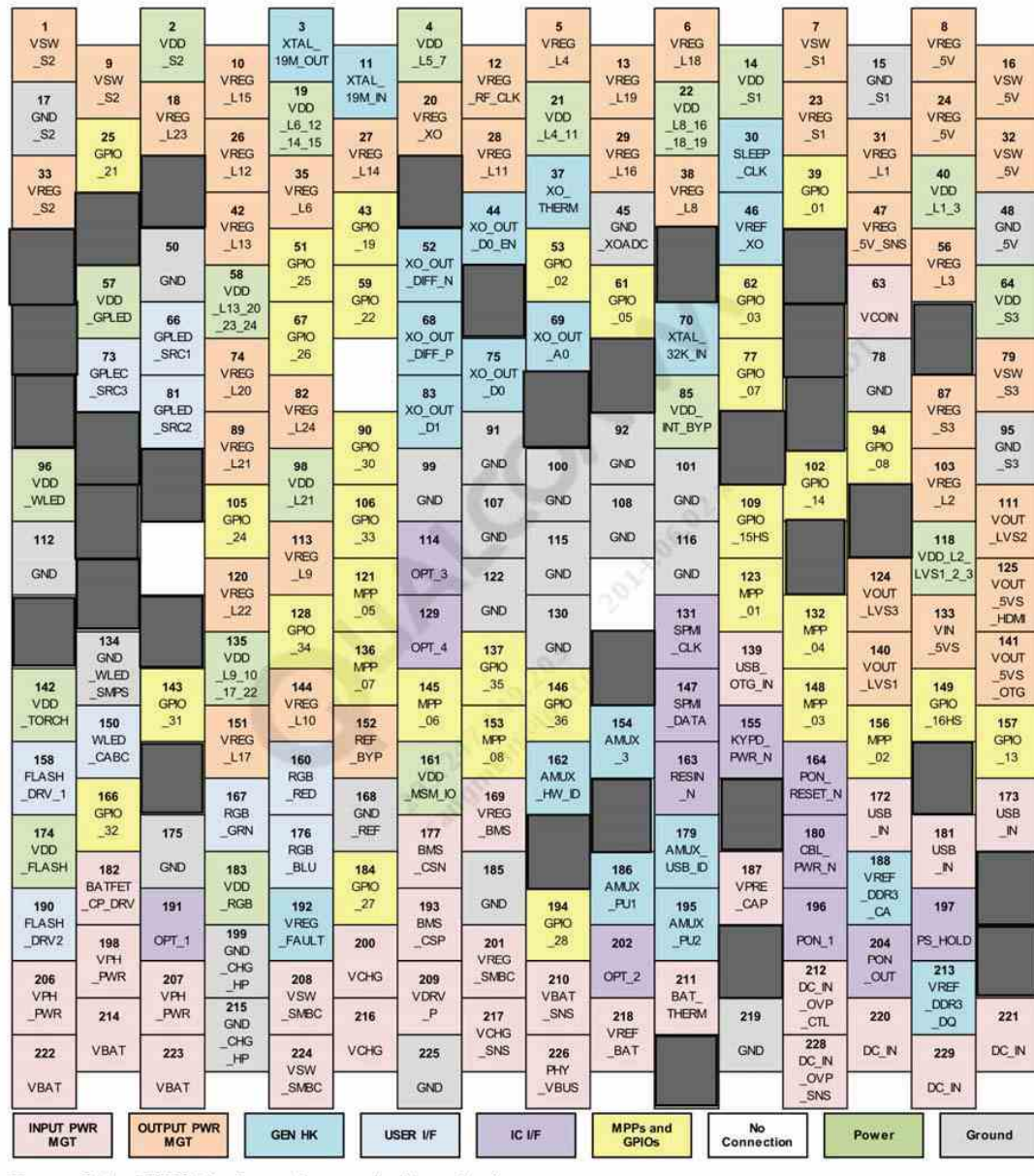
U1500 – WTR1625L (Top View)

			2 DRX_HMB4_IN		3 GND		4 DRX_HB1_IN			6 GND		7 DRX_LB4_IN		8 GND		9 PRX_HB1_IN
10 GND		11 VDD_RF1_D_MB1					14 VDD_RF1_D_LB_LO			16 DRX_LB3_IN						19 PRX_HMB4_IN
			21 GND		22 VDD_RF1_D_HB		23 GND			24 GND		25 VDD_RF1_P_HB_LO		26 GND		
28 DRX_MB1_IN									31 VDD_RF1_D_LB						34 VDD_RF1_P_HMB	35 GND
	36 GNSS_RF_INP				38 VDD_RF1_D_LOM		39 GND		40 GND			41 GND		42 GND		43 PRX_MB2_IN
44 GNSS_RF_INM		45 GND		46 GND		47 GND		48 VDD_RF1_S_VCO	49 GND							51 PRX_MB1_IN
	52 VDD_RF1_G_LNA		53 GNSS_BB_IM		54 VDD_RF2_D_BB				55 GND			56 GND		57 VDD_RF1_P_HMB_LO		58 GND
59 VDD_RF1_G_BB		60 GNSS_BB_IP		61 DRX_BB_OP				62 VDD_RF2_S_VCO	63 GND			64 GND				65 PRX_LB4_IN
	66 GND		67 GNSS_BB_OP		68 DRX_BB_OM				69 GND		70 GND	71 RTUNE		72 VDD_RF1_P_LB		73 PRX_LB3_IN
74 VDD_RF1_G_VCO		75 GND		76 DRX_BB_IP		77 GND		78 VDD_RF1_S_PLL	79 VDD_RF1_P_PLL	80 VDD_RF2_P_VCO		81 GND		82 GND		83 GND
	84 GND		85 GNSS_BB_OM		86 DRX_BB_IM		87 GND		88 GND		89 GND		90 VDD_RF1_P_VCO			92 PRX_LB2_IN
93 VDD_RF1_D_PLL		94 GND		95 SSBI_PRX_DRX		96 GND		97 PRX_BB_OM	98 VDD_RF2_P_BB		99 PRX_BB_IP		100 VDD_RF2_P_RX		101 GND	102 PRX_LB1_IN
	103 VDD_DIO		104 GP_DATAT		105 SSBI_TX_GNSS		106 GND		107 PRX_BB_OP		108 PRX_BB_IM		109 ADC_IN		110 GND	111 GND
112 GND		113 GND		114 VDD_RF2_T_PLL		115 VDD_RF1_T_SYN		116 VDD_RF2_FBRX		117 POET_RFFB		118 GND		119 GND	120 GND	121 TX_HB2_OUT
	122 GND		123 GP_DATA0		124 GND		125 GND		126 VDD_RF2_T_BB		127 DAC_REF		128 GND		129 VDD_RF2_T_DA	130 TX_HB1_OUT
131 XO_IN		132 GND		133 GND				134 GND	135 VDD_RF1_T_LO			136 VDD_RF1_T_UPC		137 VDD_RF1_T_DA	138 TX_MB2_OUT	139 TX_MB3_OUT
		140 GND		141 GND			142 GND		143 GND				144 GND	145 GND		146 TX_MB1_OUT
147 VDD_RF2_XO			148 GND		149 VDD_RF1_T_VCO				150 GND	151 TX_BB_IP		152 TX_BB_OP		153 TX_LB2_OUT	154 TX_LB4_OUT	155 TX_MB4_OUT
	156 GND		157 VDD_RF2_T_VCO		158 GND		159 GND		160 TX_BB_IM		161 TX_BB_OM		162 TX_LB1_OUT		163 TX_LB3_OUT	164 GND

 Not Used

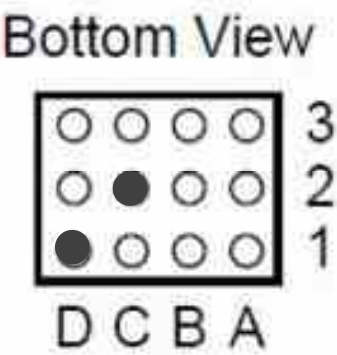
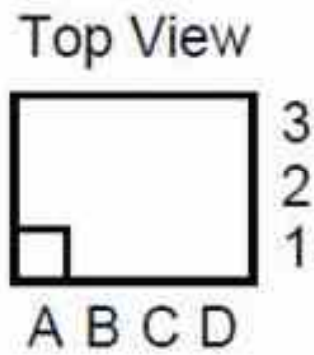
7. BGA PIN MAP

U4100-PM8941 (TOP VIEW)



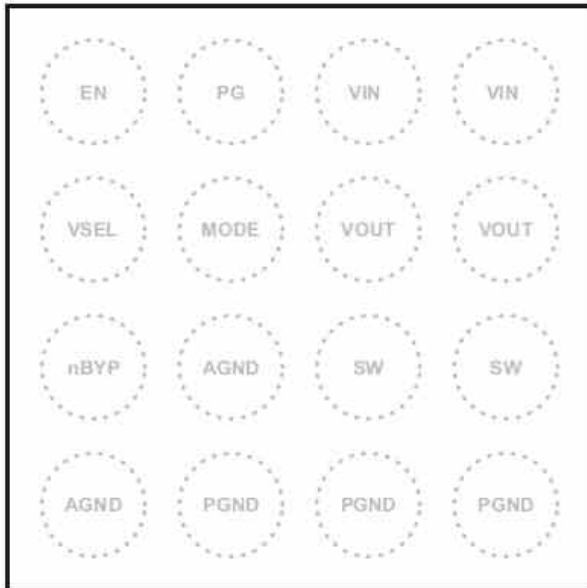
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U4400-IC,DC-DC Converter

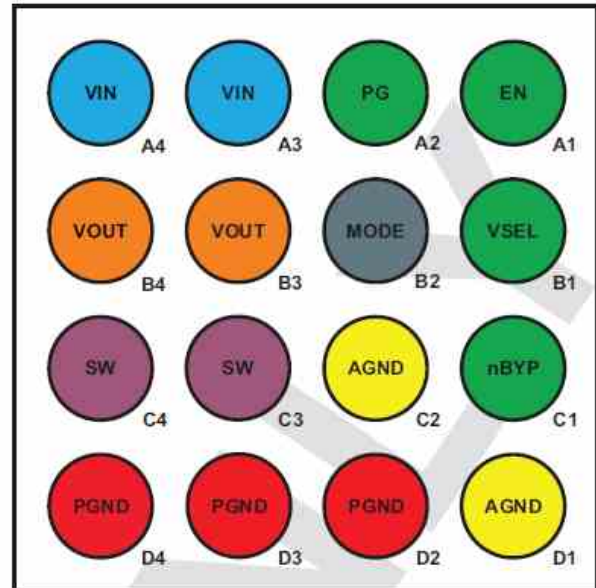


 Not Used

U4900_DC-DC Converter(TOP view)



TOP VIEW



BOTTOM VIEW

All Used

U6200_IC,Speaker Amplifier(Top view)

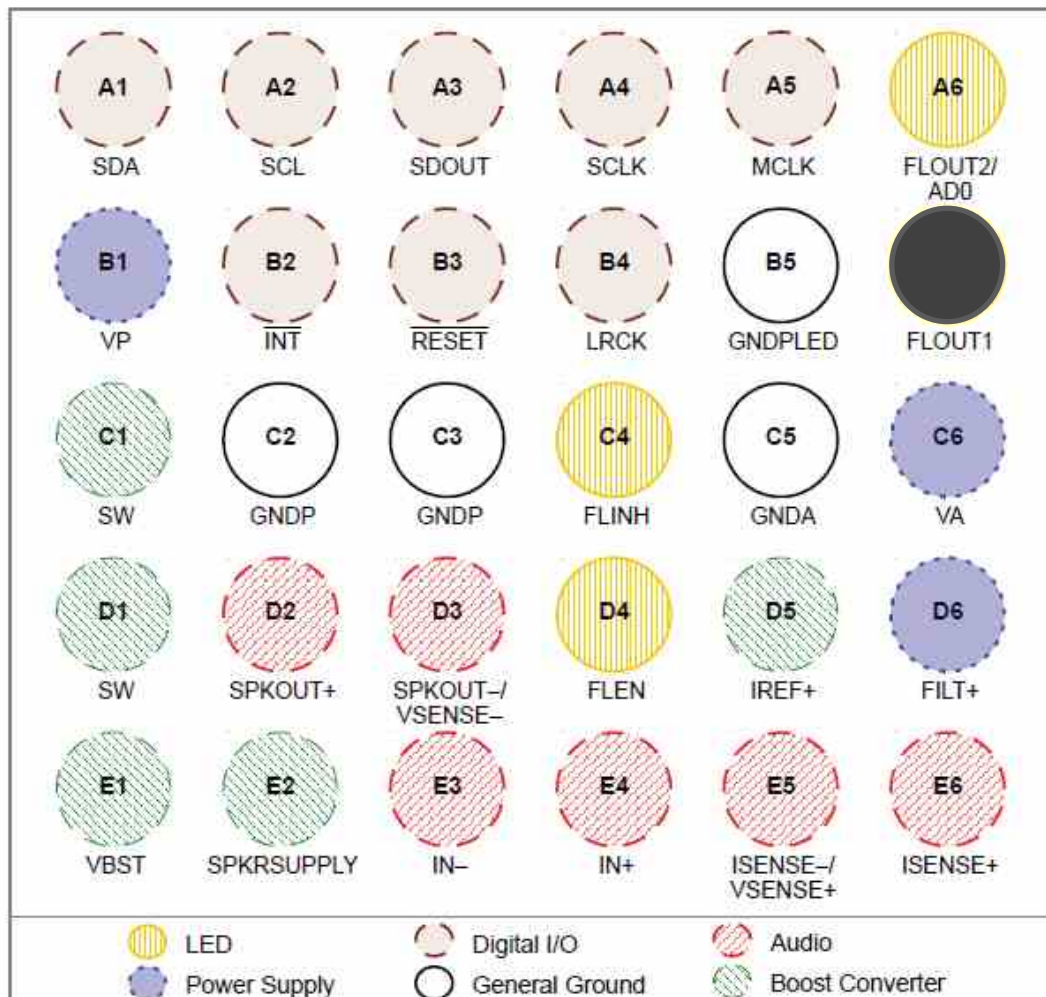
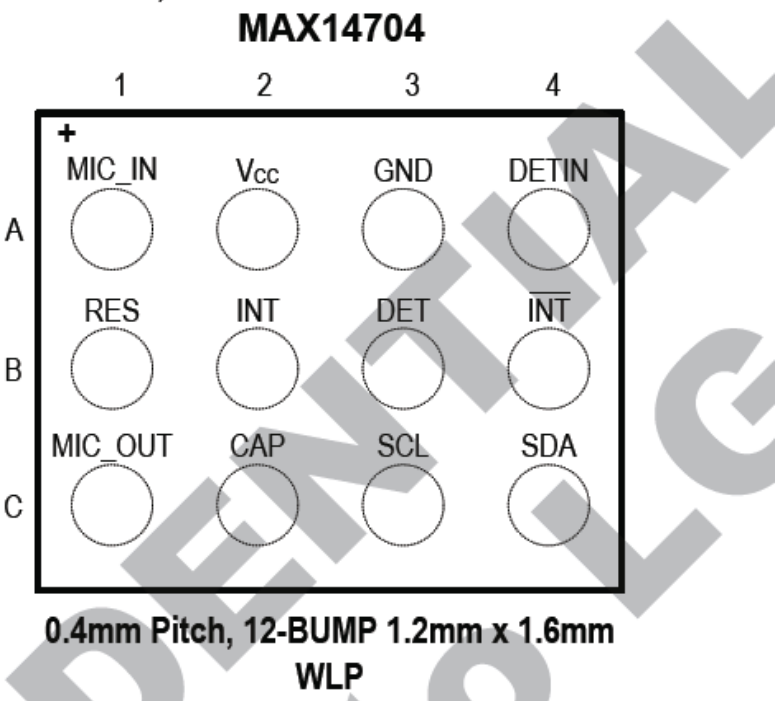


Figure 1-1. Top-Down (Through-Package) View—30-Ball WLCSP Package

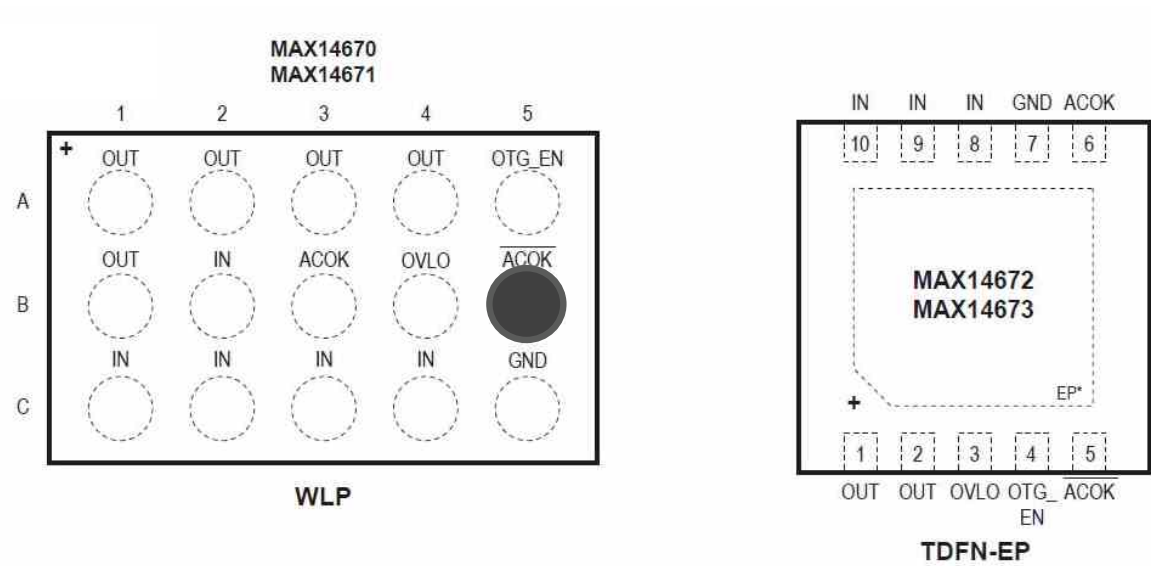
● Not Used

U6420_IC,Comparator(Top view)



ALL USED

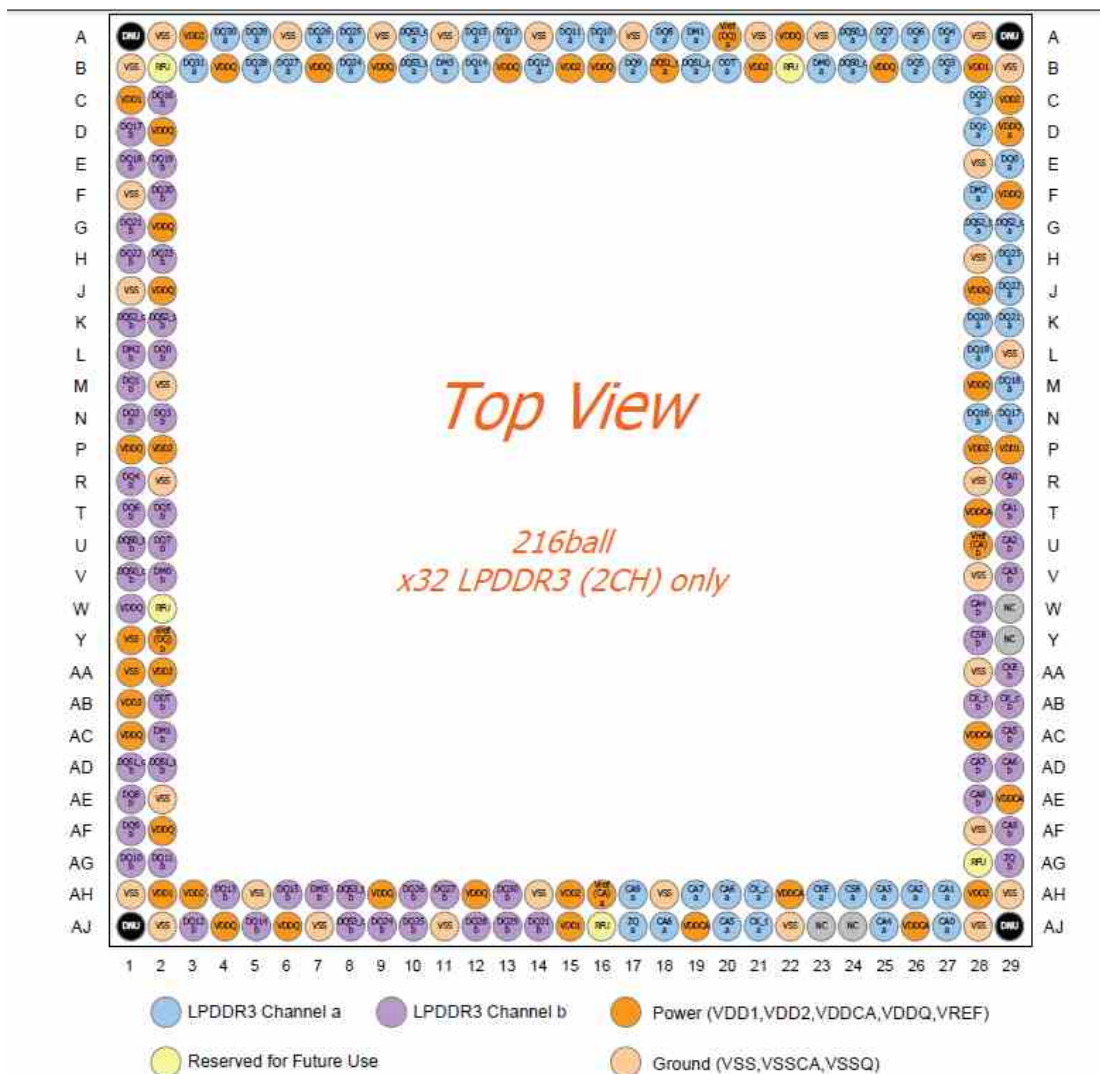
U4750,U4751_IC,over voltage protection(Top view)



Not Used

7. BGA PIN MAP

U3100_LPDDR3_SDRAM(Top View)



ALL USED